Milestone 2: how to do each section (study notes)

Working notes for the CTRS group, not part of the submission. Due **Thu 15 Oct 2026** (Course Dropbox, myCourseVille); presented with Milestone 1 on Tue 20 Oct (Sec 1) / Sat 24 Oct (Sec 5).

Sources used, cited inline:

- **Lecture notes**: `ch5 S12` = chapter 5 slide 12, in `materials/lecture_notes/chapter_0X/md/` (ch5 scope, ch6 schedule, ch7 cost, ch9 resource, ch3 the JWD case study).
- **PMBOK7**: PMBOK Guide 7th ed., cited as printed page / PDF page; see the TOC. PMBOK7 is principle-based; the course slides follow PMBOK 6 (process groups and knowledge areas), so **use the slide terminology first** and PMBOK7 for definitions.
- **Pet Care**: the 2025 sample report, `examples/md/2025_Pet_Care…md`, PDF pages 53–73 (page images in `2025_Pet_Care-…_png/page-NN.png`).
- **M1**: our Milestone 1 `part2-final-deliverables.md`.

---

0. The big picture

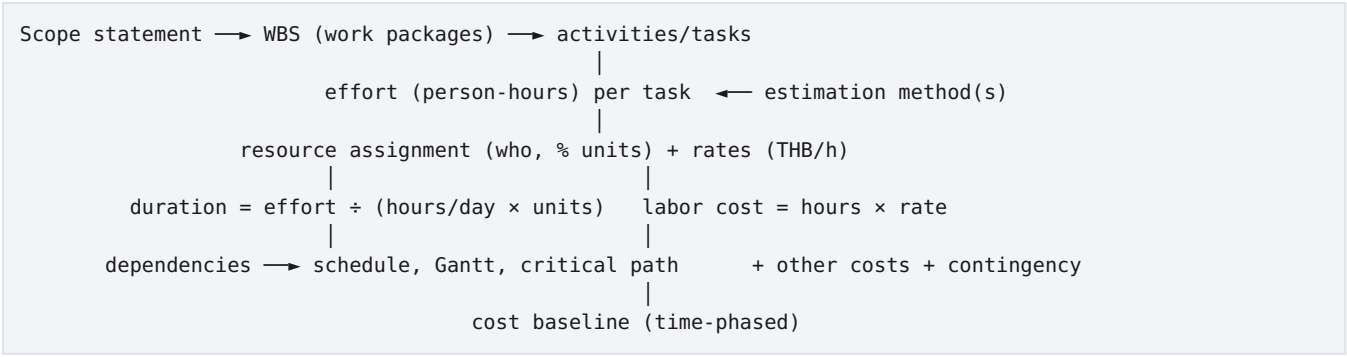
What M2 asks for (brief, Milestone 2, 8 %)

#	Deliverable	Knowledge area / slides
Part 1	GenAI Log: **2 prompt records per member** (8 records), one per page	brief "GenAI Log file"
1	Review Board Decision Log	brief
2	Project scope statement: project summary, product characteristics and requirements, deliverables (PM-related + product-related), exclusions, success criteria	Scope, ch5
3	WBS and tasks	Scope, ch5; activities ch6
4	Effort estimation for tasks	Cost ch7, Schedule ch6
5	Human and physical resource assignment to tasks, resource rates	Resource ch9
6	Cost estimation: labor + other costs + contingency reserve, **aligned with assignment, rates and effort**	Cost ch7
7	Schedule from WBS/tasks with resources: **Gantt chart view and critical path**	Schedule ch6

Scoring: PMBOK correctness, **consistency between artifacts**, quality of critical reasoning in prompting, presentation. Deliverables must be "correct and complete but CONCISE".

One chain, built in this order

The seven items are one model. Every number flows from the WBS, so build them in this order (this is exactly what the JWD case does, ch3 S23–24: scope statement and WBS together, then ask each member which tasks they do and how many hours a month they can give, then dependencies, hours per task, durations, critical path, adjust to the charter's dates):



Do it **in a scheduling tool** (ProjectLibre or GanttProject, both listed in the syllabus §15.4): enter WBS + tasks + dependencies + resources with rates + work (hours) once, and the tool gives you durations, Gantt, critical path, resource usage and cost per task. Export tables from the same file, so items 3–7 can't contradict each other. This is the single most important lesson from Pet Care, whose numbers disagree across tables (see §8).

Anchors from M1 that M2 must stay consistent with

Item	M1 value (where)
Dates	3 Aug 2026 to 28 Jul 2028; iterations I1, I2, E1, E2, C1, C2, T1, C3, T2, P1, C4, T3, close-out (Table 2.21)
Milestones	LCO 2 Sep 2026 · LCA 23 Oct 2026 · IOC 26 Feb 2027 · R1.0 in operation 2 Apr 2027 · R2.0 30 Jul 2027 · R2.1 11 Feb 2028 · expansion decision / close 28 Jul 2028
Scope	Releases 1.0 (F1–F28), 2.0 (F29–F35), 2.1 productization; Future F36, F37 out of scope (Tables 2.13–2.18)
Quality targets	5 TPS, 2 s response, 15 s check-in, 2 h recovery, browsers, daily backup (Table 2.19)
Team and rates	Boonnisa T. PM 350 THB/h · Sirapatr C. SA 350 · Chawanakorn S. Req. Specifier/Designer 300 · Watayut A. QA/ Tester 300 (Table 3.4)
Effort (top-down)	2,632 h total: Year 0 2,092 h, Year 1 540 h, split by phase and person (Table 3.4)
Budget	1,010,590 THB = Year 0 767,690 (labor 674,600 + devices 15,000 + hosting/domain 8,300 + contingency 69,790) + Year 1 242,900 (labor 172,000 + legal 15,000 + hosting/support 37,200 + contingency 18,700) (Tables 2.20, 3.7)
Risks	RISK-01 … RISK-14 (Table 2.24), which justify the contingency reserve
Success criteria	7 criteria in the charter (Table 4.1)

M1 itself promises: "Milestone 2 replaces this top-down figure by a bottom-up estimate per WBS task" (§3.3.1). So M2's headline is: **bottom-up effort and cost per WBS task, compared with the M1 top-down estimate**. If the totals change, say so: add a row to the Document Revision History, record the decision in the Review Board log, and update the budget/charter figure or explain the variance.

---

1. Review Board Decision Log

Format (brief): `No. | Artifact | Member (prompt design) | Suggestion from team (in brief) | Decision (Accept / Revise / Reject)`. Same as M1 Table 1.1; copy its style (one row per artifact or artifact part, a short concrete suggestion, a decision).

How to do it:

1. After each member's prompting, hold one Review Board meeting (all four) per artifact set.
2. Review each artifact against the three evaluation criteria of the brief: **Requirement Satisfaction**, **PMBOK Correctness**, **Project Consistency** (with M1 and with the other M2 artifacts).
3. Record concrete suggestions, e.g. "Effort for C1 payment tasks too low compared with M1 Table 3.4", "WBS mixes nouns and verbs", "Critical path list includes tasks with slack".
4. Make sure the member names match the GenAI log records (each member ≥ 2 prompts).

Pet Care has **no** decision log or GenAI section at all; don't use it as a model for this part.

---

2. Project scope statement

Definitions

- **Scope** = "all the work involved in creating the products of the project and the processes used to create them" (ch5 S4). Scope management = "defining and controlling **what is or is not included**" (ch5 S4); "all the work required, and only the work required" (ch5 S33).
- **Project scope statement** = "The description of the project scope, major deliverables, and **exclusions**" (PMBOK7 glossary, p.236 / PDF 341). It identifies "the major deliverables ... and the **acceptance criteria for each deliverable**" (PMBOK7 p.84 / PDF 179).
- **Product scope** = features and functions of the product; **project scope** = the work to deliver it (PMBOK7 p.54 / PDF 149).
- Deliverables are **product related** (e.g. software) or **process related** (e.g. planning documents, meeting minutes) (ch5 S4). These are the brief's "product-related" and "project management-related" deliverables.

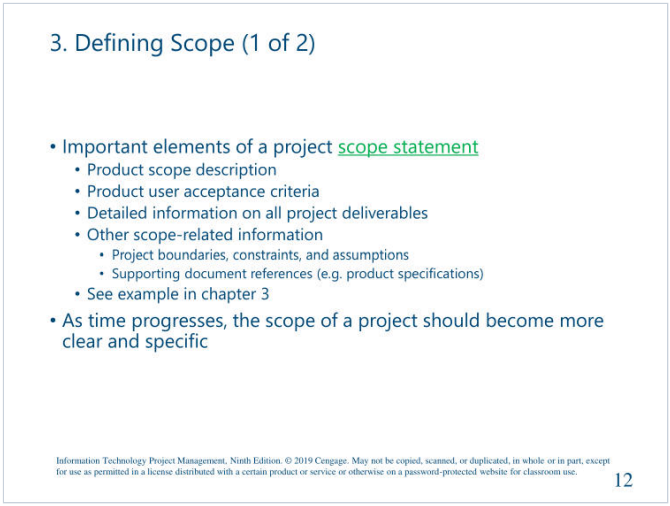


Figure 1. ch5 S12 · Contents of a project scope statement

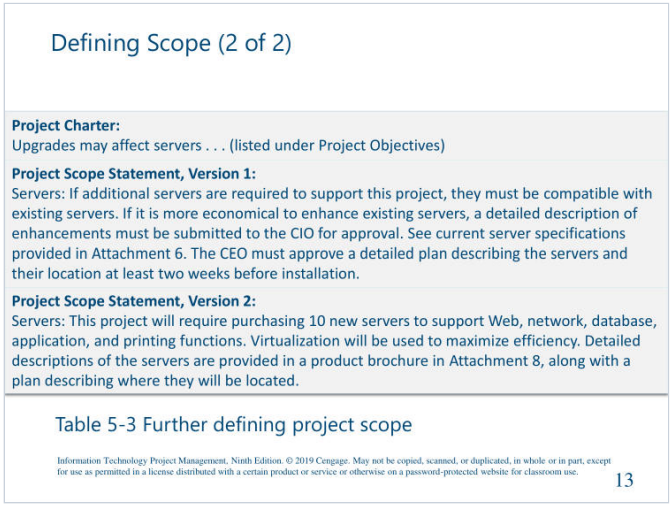


Figure 2. ch5 S13 · Progressive elaboration: vague vs specific scope (Table 5-3)

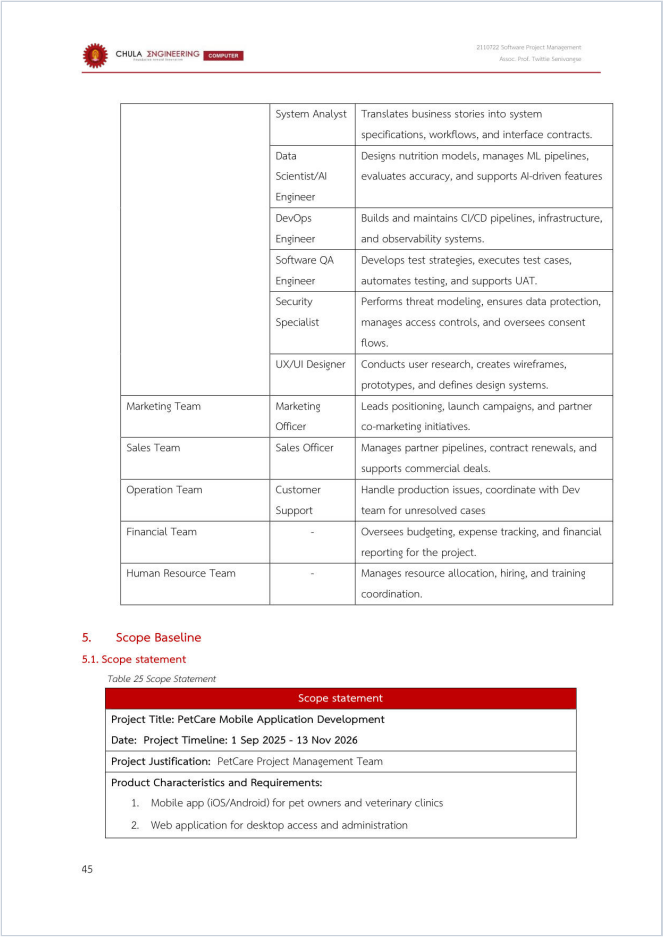
Contents (ch5 S12 mapped to the brief's headings)

Brief heading	Slide content (ch5 S12)	What to write for CTRS
Project summary	(Project title, date, justification)	Title, date, 3–5 sentence summary: La Loy pilot, LINE self-service reservation, two releases + productization, dates, budget (from charter). Don't leave it as filler: Pet Care's "Project Justification" just names the team.
Product characteristics and requirements	"Product scope description"	Releases 1.0 / 2.0 / 2.1 and the feature list by role (summarize M1 Tables 2.15–2.18 by feature ID; don't paste the whole tables again), quality targets of Table 2.19, legal (PDPA, Alcoholic Beverage Control Act).
Project deliverables: PM-related	"Detailed information on all project deliverables"	Business case, charter, scope statement, WBS + dictionary, schedule/cost baselines, quality/risk/communication plans, iteration plans and assessments, status reports, milestone review records (LCO, LCA, IOC, PR), pilot KPI reports, expansion decision record, lessons learned (M1 Table 2.23 lists them by phase).
Project deliverables: product-related	(same)	RUP artifacts (Vision, Use-Case Model, SRS, Supplementary Spec, SAD, UI prototype, test plan and cases), working builds per iteration, Release 1.0, 2.0, 2.1 in operation, user training material, data migration, deployment/hosting setup, operations/support routine.
What is excluded	"Project boundaries, constraints, and assumptions"	**Explicit list**: F36 split payment, F37 table photos, configurable business rules (Future); the restaurant's check-in tablet (La Loy buys it); LINE OA plan and gateway merchant fees (restaurant's operating cost); sales and onboarding of other restaurants after the gate (Years 2–5 operation, outside project); hardware at the venue; refunds policy decisions (the Owner's). Add the key assumptions and constraints (cite ASM/CON IDs from M1 §2.4).

Brief heading	Slide content (ch5 S12)	What to write for CTRS
Project success criteria	"Product user acceptance criteria"	Reuse the 7 charter success criteria **verbatim or by reference** plus acceptance criteria per major deliverable (e.g. "Release 1.0 accepted when acceptance tests of the two main use cases pass with the Manager and Front Staff"). Must be measurable.
(optional)	"Supporting document references"	Vision v1.x, SRS, Development Case in the 2110628 repo.

Tips:

- Be **specific**: ch5 S13 contrasts a vague scope version ("if additional servers are required…") with a specific one ("purchase 10 new servers … details in Attachment 8"). Name feature IDs, dates and numbers.
- **Format**: Pet Care used a 2-column labelled table (Table 25). Copy that, but remove template leftovers (their product deliverables still say "research reports, design documents, software code, hardware, etc.").
- The scope statement + WBS + WBS dictionary = **scope baseline** (ch5 S14; PMBOK7 glossary p.239 / PDF 344).



CHULA ENGINEERING COMPUTER

2110722 Software Project Management
Assoc. Prof. Twittie Senivongse

	System Analyst	Translates business stories into system specifications, workflows, and interface contracts.
	Data Scientist/AI Engineer	Designs nutrition models, manages ML pipelines, evaluates accuracy, and supports AI-driven features
	DevOps Engineer	Builds and maintains CI/CD pipelines, infrastructure, and observability systems.
	Software QA Engineer	Develops test strategies, executes test cases, automates testing, and supports UAT.
	Security Specialist	Performs threat modeling, ensures data protection, manages access controls, and oversees consent flows.
	UX/UI Designer	Conducts user research, creates wireframes, prototypes, and defines design systems.
Marketing Team	Marketing Officer	Leads positioning, launch campaigns, and partner co-marketing initiatives.
Sales Team	Sales Officer	Manages partner pipelines, contract renewals, and supports commercial deals.
Operation Team	Customer Support	Handle production issues, coordinate with Dev team for unresolved cases
Financial Team	-	Oversees budgeting, expense tracking, and financial reporting for the project.
Human Resource Team	-	Manages resource allocation, hiring, and training coordination.

5. Scope Baseline

5.1. Scope statement

Table 25 Scope Statement

Scope statement
Project Title: PetCare Mobile Application Development **Date: Project Timeline: 1 Sep 2025 - 13 Nov 2026**
Project Justification: PetCare Project Management Team
Product Characteristics and Requirements: 1. Mobile app (iOS/Android) for pet owners and veterinary clinics 2. Web application for desktop access and administration

45

Figure 3. Pet Care p.53 · Scope statement as a labelled table (Table 25)

3. WBS and tasks

Definitions and rules

- **WBS** = "a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project", the "foundation document … for planning and managing project schedules, costs, resources, and changes" (ch5 S14). PMBOK7: "hierarchical decomposition of the total scope of work" (p.243 / PDF 348).
- **Work package** = "a task at the lowest level of the WBS" (ch5 S14); "for which cost and duration are estimated and managed" (PMBOK7 p.243 / PDF 348).
- "PMI recommends using **nouns** to name WBS items" (ch5 S14); activities below them are **verbs** (ch6 S8).
- Rules (ch5 S26), the course's version of the 100 % rule:
- a unit of work appears in **only one place**;
- the work content of a WBS item is the **sum** of the items below it;
- each WBS item is the responsibility of **only one individual**;
- the WBS must match how the work is actually going to be performed;
- team members help build it;
- each WBS item is documented in a **WBS dictionary**.
- Four approaches (ch5 S23): guidelines, analogy (tailor a similar project's WBS, e.g. Pet Care's), top-down, bottom-up.
- Ways to organize: by **product** (ch5 S15), by **phase** with a Project Management branch (ch5 S16–17: 1.1 Project Management, 1.2 Product Requirements, 1.3 Detail Design, 1.4 Construct, 1.5 Integration and Test), or by **process group** (ch5 S20, ch3 S24 "may use the process groups as the categories of work"). Agile: by process (ch5 S21, sprint tasks with Duration and Dependency) or by product (ch5 S22, product backlog = WBS: `EPIC ID | EPIC Name | User Story ID | User Story | Estimate (SP)`).

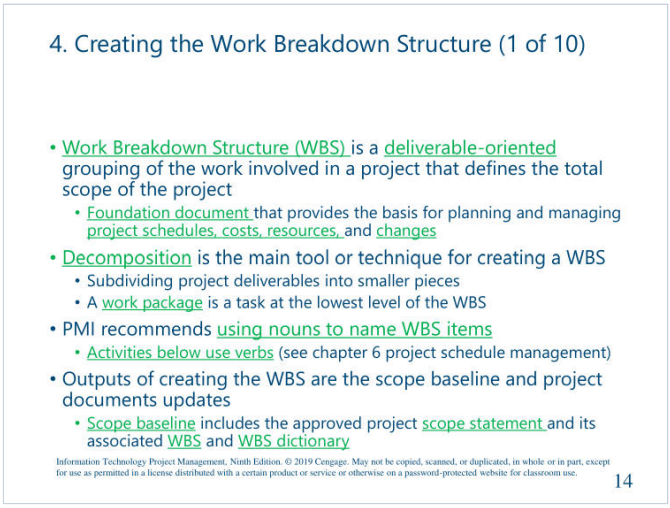


Figure 4. ch5 S14 · WBS definition, work package, scope baseline

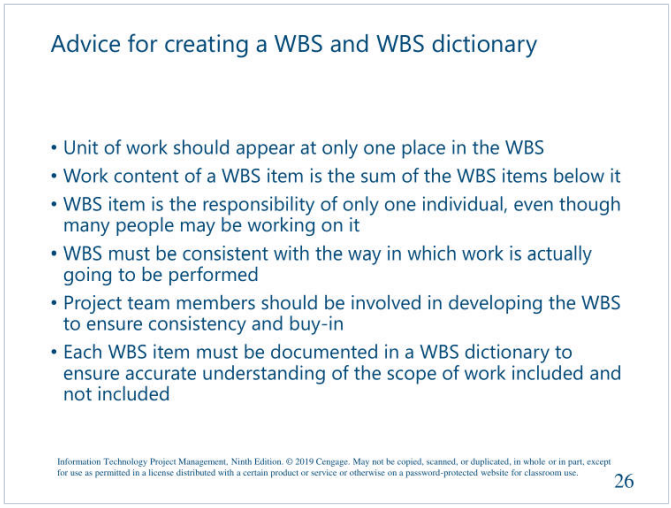


Figure 5. ch5 S26 · Advice for creating a WBS and WBS dictionary

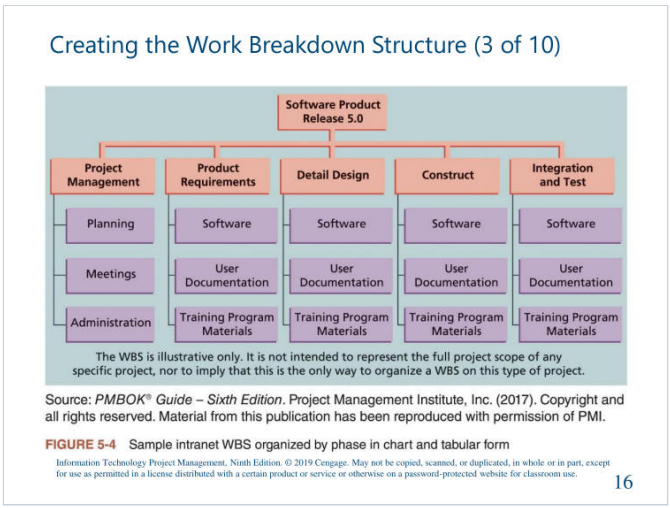


Figure 6. ch5 S16 · WBS organized by phase (Figure 5-4)

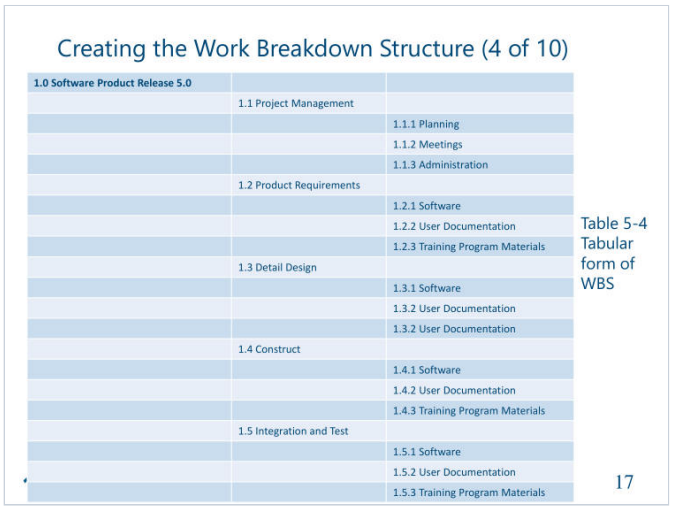


Figure 7. ch5 S17 · Same WBS in tabular form (Table 5-4)

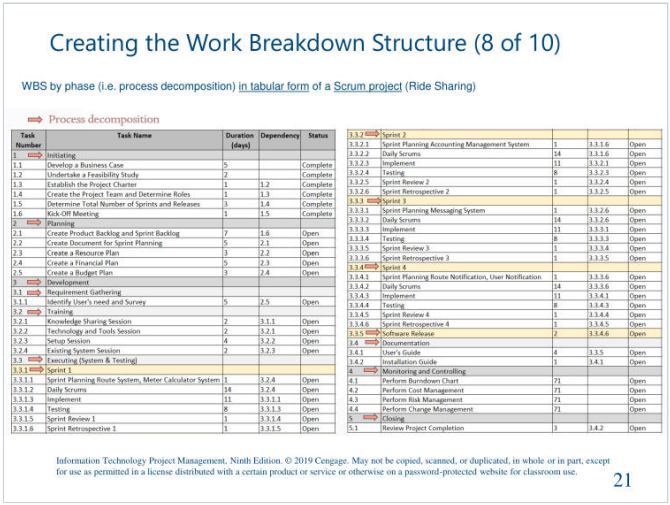


Figure 8. ch5 S21 · Agile WBS by process: sprint tasks with duration and dependency

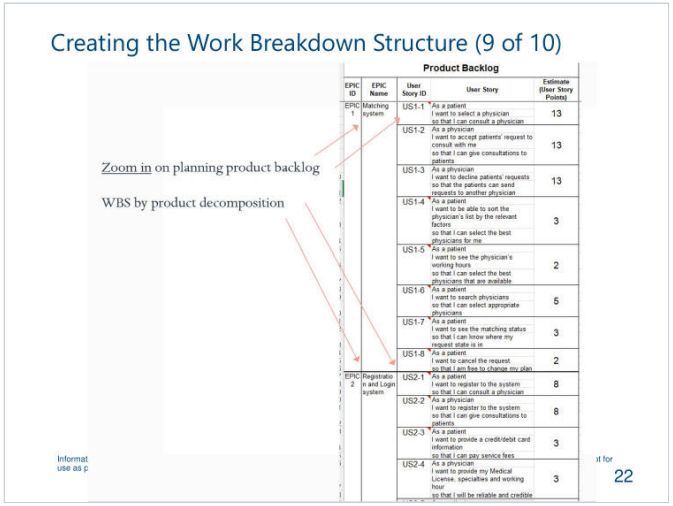


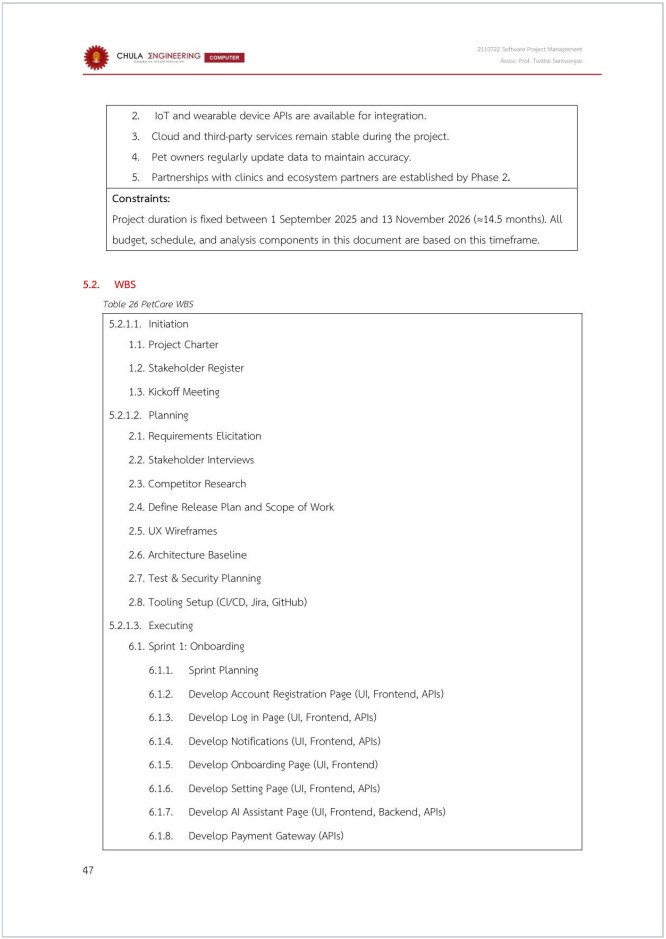
Figure 9. ch5 S22 · Agile WBS by product: product backlog with story points

Suggested WBS for CTRS (tailored RUP, phase/iteration then deliverable)

Level 1 project, level 2 phase (plus a PM branch), level 3 iteration, level 4 work packages (nouns); activities/tasks under them carry the effort. Keep **one numbering scheme** and use it in every table and in the tool.

```
1 CTRS Project
1.1 Project Management                (process-related, runs throughout)
    1.1.1 Initiation: business case, charter, organization (M1)
    1.1.2 Planning: scope statement, WBS, estimates, schedule & cost baseline (M2), quality/risk/
communication plans (M3)
    1.1.3 Monitoring and control: weekly status assessments, iteration assessments, CCB, risk reviews
    1.1.4 Milestone reviews: LCO, LCA, IOC, Product Release 1.0 / 2.0 / 2.1
    1.1.5 Closing: pilot review, expansion decision, lessons learned, sign-off
1.2 Inception (I1, I2)       → Vision, Stakeholder Requests, Glossary, Development Case, iteration plans,
Risk List
1.3 Elaboration (E1, E2)     → use-case specs, SRS, Supplementary Spec, Analysis Model, SAD, UI prototype,
test plan, payment/LINE sandbox
1.4 Release 1.0 Core
    1.4.1 Construction C1    → work packages per feature group: concert set-up (F10, F11), table map and
hold (F2, F3, F21), payment (F4, F23, F27), staff accounts (F24)…
    1.4.2 Construction C2    → e-ticket and messages (F5, F6), check-in (F15–F20), live view (F12),
exceptions (F13, F14), audit/backup/consent (F22, F25, F26, F28), load test
    1.4.3 Transition T1      → pilot nights, training, data migration, go-live
1.5 Release 2.0 Growth (C3, T2)  → F29–F35, training, go-live
1.6 Pilot operation (P1)     → concert-night support, monthly KPI reports
1.7 Release 2.1 Product (C4, T3) → multi-tenant data model, plans and limits, billing, onboarding, service
admin, La Loy migration
```

Each Construction work package can be decomposed into the same activities: design → implement → unit test → integrate → system test. That is what makes bottom-up estimating possible (§4). In Pet Care every development task was "Develop X Page (UI, Frontend, Backend, APIs, Databases)", one task per page per sprint; they had ~100 leaf tasks and 116 Gantt rows. **Aim for a WBS deep enough to estimate (roughly 60–120 tasks) but present a summarized version** if it's long (brief: concise).



CHULA ENGINEERING COMPUTER

2110722 Software Project Management
Assoc. Prof. Twittie Senivongse

2. IoT and wearable device APIs are available for integration.
3. Cloud and third-party services remain stable during the project.
4. Pet owners regularly update data to maintain accuracy.
5. Partnerships with clinics and ecosystem partners are established by Phase 2.

Constraints:
Project duration is fixed between 1 September 2025 and 13 November 2026 (≈14.5 months). All budget, schedule, and analysis components in this document are based on this timeframe.

5.2. WBS

Table 26 PetCore WBS

5.2.1.1. Initiation
1.1. Project Charter
1.2. Stakeholder Register
1.3. Kickoff Meeting
5.2.1.2. Planning
2.1. Requirements Elicitation
2.2. Stakeholder Interviews
2.3. Competitor Research
2.4. Define Release Plan and Scope of Work
2.5. UX Wireframes
2.6. Architecture Baseline
2.7. Test & Security Planning
2.8. Tooling Setup (CI/CD, Jira, GitHub)
5.2.1.3. Executing
6.1. Sprint 1: Onboarding
6.1.1. Sprint Planning
6.1.2. Develop Account Registration Page (UI, Frontend, APIs)
6.1.3. Develop Log in Page (UI, Frontend, APIs)
6.1.4. Develop Notifications (UI, Frontend, APIs)
6.1.5. Develop Onboarding Page (UI, Frontend)
6.1.6. Develop Setting Page (UI, Frontend, APIs)
6.1.7. Develop AI Assistant Page (UI, Frontend, Backend, APIs)
6.1.8. Develop Payment Gateway (APIs)

47

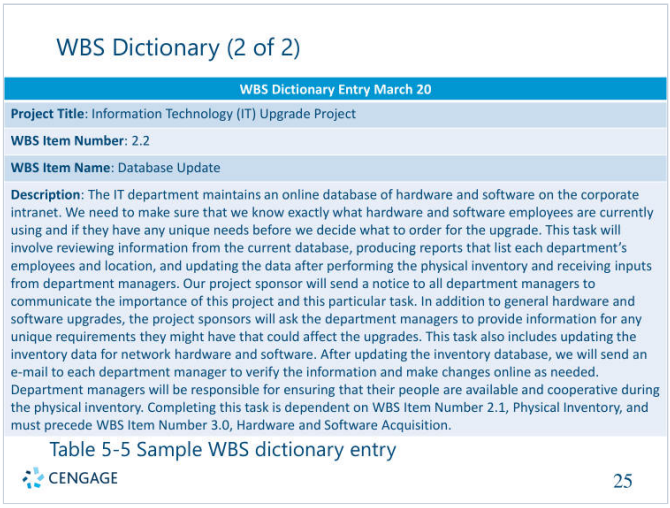
Figure 10. Pet Care p.55 · WBS as an indented outline (Table 26); note the numbering

Tasks / activity list

- **Activity list** = "Activity name, activity identifier or number, and brief description"; **activity attributes** = "predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, assumptions" (ch6 S8).
- **Milestones**: zero duration, e.g. customer sign-off (ch6 S9); make them SMART (ch6 S22). Put LCO, LCA, IOC, R1.0, R2.0, R2.1, close-out as 0-day milestones.

WBS dictionary

- "Describes detailed information about each WBS item ... many WBS tasks are vague" (ch5 S24–25). Template (Table 5-5): **Project Title, Date, WBS Item Number, WBS Item Name, Description** (what work, who is responsible, dependencies such as "dependent on 2.1 ... must precede 3.0").
- Pet Care fields (Tables 27–28), worth copying: **Project Name, WBS Item Number, WBS Item Name, Description, Responsibility, Input, Output, Acceptance Criteria**. Improve on them by adding **Effort (h), Duration, Cost, Predecessors**, and pick at least one **product** work package (they only did two PM items), e.g. "1.4.1.3 Gateway payment" and "1.1.3 Monitoring and control".



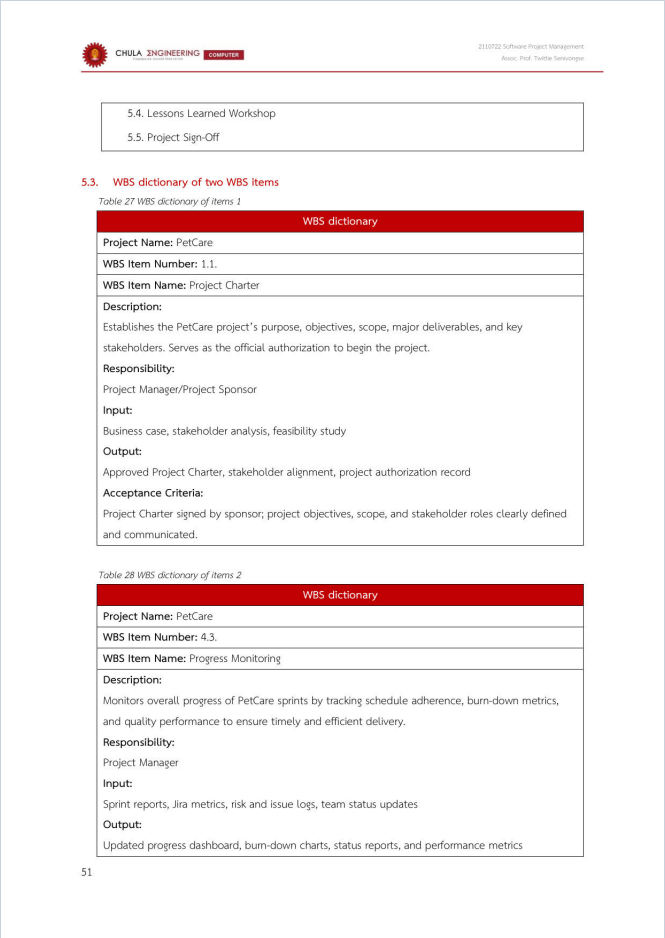
WBS Dictionary (2 of 2)

WBS Dictionary Entry March 20
Project Title: Information Technology (IT) Upgrade Project
WBS Item Number: 2.2
WBS Item Name: Database Update
Description: The IT department maintains an online database of hardware and software on the corporate intranet. We need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Physical Inventory, and must precede WBS Item Number 3.0, Hardware and Software Acquisition.

Table 5-5 Sample WBS dictionary entry

CENGAGE 25

Figure 11. ch5 S25 · WBS dictionary template (Table 5-5)



CHULA ENGINEERING COMPUTER

5.4. Lessons Learned Workshop

5.5. Project Sign-Off

5.3. WBS dictionary of two WBS items

Table 27 WBS dictionary of items 1

WBS dictionary
Project Name: PetCare
WBS Item Number: 1.1.
WBS Item Name: Project Charter
Description: Establishes the PetCare project's purpose, objectives, scope, major deliverables, and key stakeholders. Serves as the official authorization to begin the project. **Responsibility:** Project Manager/Project Sponsor **Input:** Business case, stakeholder analysis, feasibility study **Output:** Approved Project Charter, stakeholder alignment, project authorization record **Acceptance Criteria:** Project Charter signed by sponsor; project objectives, scope, and stakeholder roles clearly defined and communicated.

Table 28 WBS dictionary of items 2

WBS dictionary
Project Name: PetCare
WBS Item Number: 4.3.
WBS Item Name: Progress Monitoring
Description: Monitors overall progress of PetCare sprints by tracking schedule adherence, burn-down metrics, and quality performance to ensure timely and efficient delivery. **Responsibility:** Project Manager **Input:** Sprint reports, Jira metrics, risk and issue logs, team status updates **Output:** Updated progress dashboard, burn-down charts, status reports, and performance metrics

51

Figure 12. Pet Care p.59 · WBS dictionary entries (Tables 27–28)

4. Effort estimation for tasks

Concepts

- **Effort ≠ duration**: effort = labor units (person-hours); duration = work periods. "20 h of effort may be spread over 2 months" (ch6 S15; PMBOK7 glossary PDF 334). Estimate **effort per task in person-hours**; the tool derives duration from the resource's availability.
- Absolute estimate example: "120 hours of work. One person working full time could accomplish the work in 15 workdays, assuming 8 hours of productivity per workday" (PMBOK7 p.57 / PDF 152).
- Estimate types and accuracy (ch7 S11, Table 7-1): ROM −50 % to +100 %; budgetary −10 % to +25 %; **definitive −5 % to +10 %**. M1 was ROM/budgetary top-down; M2 per task is closer to definitive.
- State a **basis of estimates**: assumptions (rates, productivity, hours a week) and the technique used (ch7 S12).
- People doing the work should estimate; an expert reviews; use similar projects; consider resource availability (ch6 S15). Watch the biases of ch7 S14: estimating too fast, underestimation, management pressure.

Types of Cost Estimates (1 of 2)

Type of Estimate	When Done	Why Done	Typical Range of accuracy
Rough order of magnitude (ROM)	Very early in the project life cycle, often 3+ years before project completion	Provides estimate of cost for selection decisions	-50% to + 100%
Budgetary	Early, 1–2 years before project completion	Puts amount in the budget plans	-10% to +25%
Definitive	Later in the project, 1 year or less before project completion	Provides details for making purchasing decisions, estimates actual costs	-5% to +10%

Table 7-1 Types of cost estimates

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

11

Figure 13. ch7 S11 · Types of cost estimates and accuracy (Table 7-1)

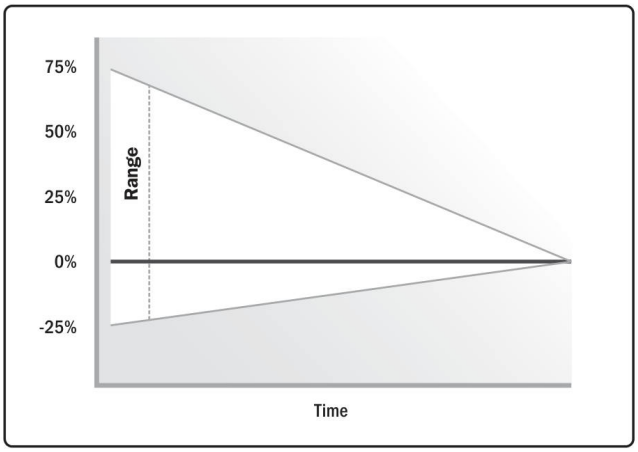


Figure 2-14. Estimate Range Decreases over Time

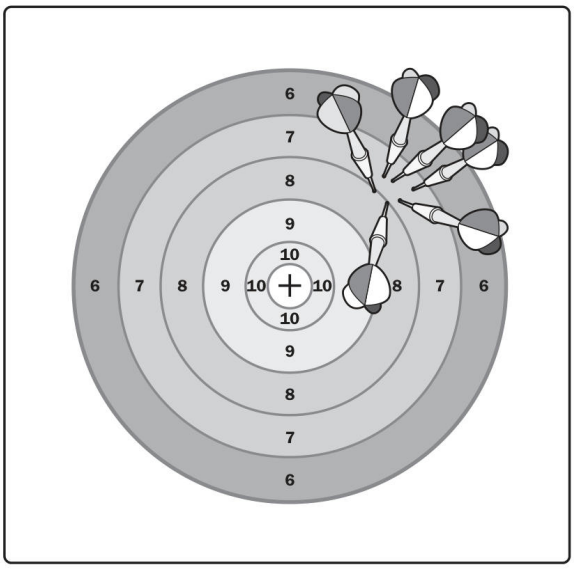


Figure 2-15. Low Accuracy, High Precision

Figure 14. PMBOK7 Figures 2-14 and 2-15 · Estimate range over time; accuracy vs precision (p.55–56)

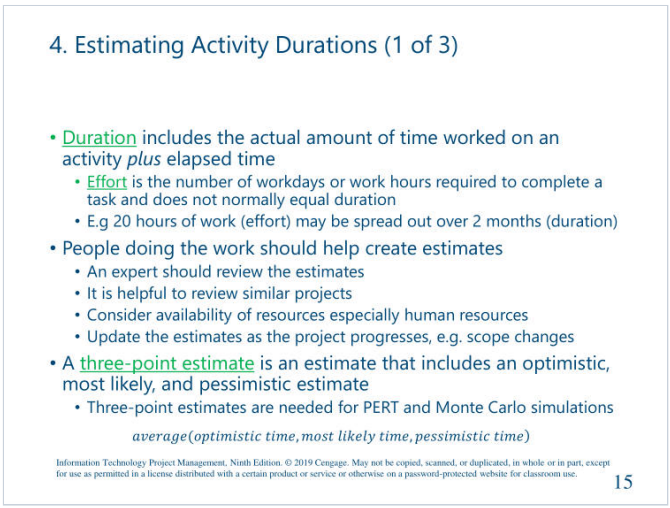


Figure 15. ch6 S15 · Effort vs duration; three-point estimates

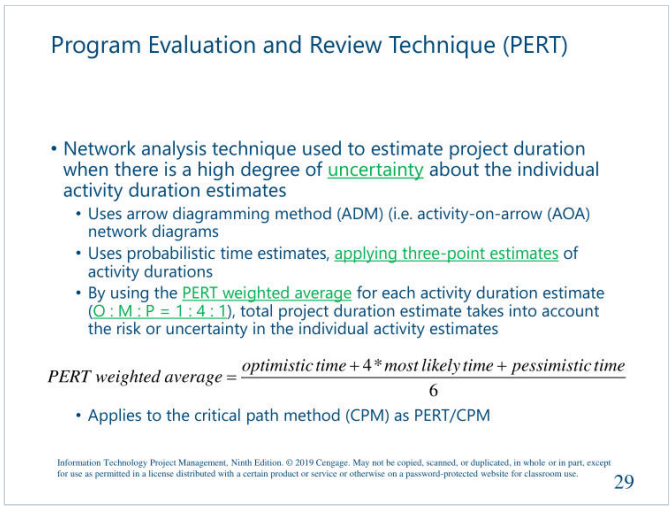


Figure 16. ch6 S29 · PERT weighted average (O + 4M + P) / 6

Techniques (ch7 S13, ch6 S5; PMBOK7 4.4.2 p.178 / PDF 273)

Technique	How	Good for CTRS
Bottom-up	estimate each task, sum up	main method, per WBS task
Analogous	actuals of a similar project	Pet Care sprint lengths, our RE course hours
Expert judgment	team's experience	PM and documentation tasks
Three-point / PERT	O, M, P; PERT = **(O + 4M + P) / 6** (ch6 S29); simple average (O+M+P)/3 (ch6 S15)	risky tasks: payment gateway, LINE integration, load test, multi-tenancy
Parametric	model from parameters: **function points**, COCOMO II, LOC	cross-check of Construction
User story points	reference story (e.g. 3 SP), relative sizing with Planning Poker 1, 2, 3, 5, 8, 13, 20…; velocity → number of increments (ch7 S22–23)	cross-check per feature F1–F35

Instructor expectation (ch7 S15): use two approaches and compare. `percent difference = |A − B| / average(A, B) × 100`. Surveyor Pro example: labor $594,000 vs FP $567,000, difference 4.65 % (ch7 S16–17).

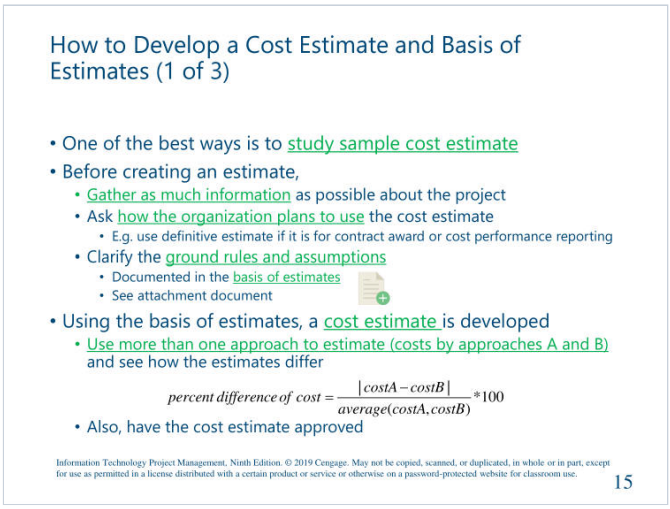


Figure 17. ch7 S15 · Use two approaches; percent difference

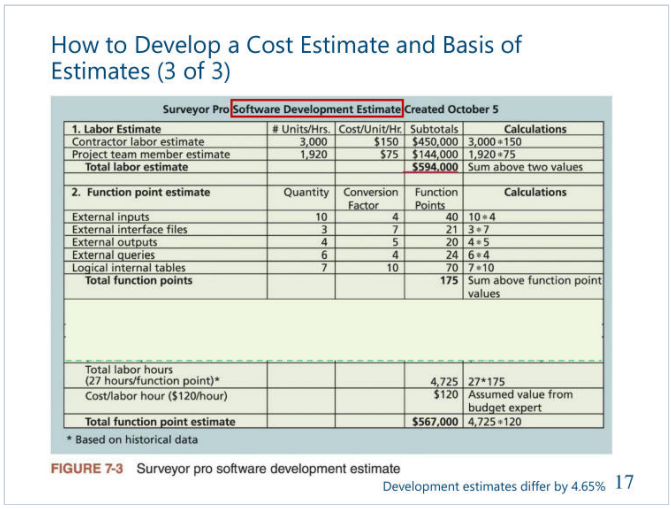


Figure 18. ch7 S17 · Labor vs function-point estimate of the same work (Figure 7-3)

Recommended for CTRS

1. **Bottom-up (approach A)**: hours per task in the WBS, estimated by the member who will do it; three-point for the risky ones. Sum per phase and per person and **compare with M1 Table 3.4** (2,632 h). That comparison is the "two approaches" check, plus it validates M1.

2. **Parametric (approach B)** for the Construction work: either - **Function points** (ch7 S24–46): count EI/EO/EQ/ILF/EIF of Release 1.0 (from the use cases), weights from the complexity matrices (e.g. EI Low 3 / Avg 4 / High 6; EO 4/5/7; EQ 3/4/6; ILF 7/10/15; EIF 5/7/10), optional VAF = 0.65 + 0.01 × ΣGSC; effort = FP × hours/FP (example 27 h/FP, or 0.25 FP per person-hour). Optional FP→LOC with QSM gearing factors (JavaScript 47, Java 53, C# 54, SQL 21 LOC/FP) and then COCOMO II: PM = 2.94 × KSLOC^E × ΠEM, E = 0.91 + 0.01 ΣSF, 1 PM = 152 h (ch7 S47–52); or - **Story points**: size F1–F35 relative to a reference story, assume a velocity per iteration, derive iterations and hours (ch7 S22–23).
3. Show the result as a table, e.g.

WBS ID	Task	Method	O	M	P	Effort (h)	Resource(s)
1.4.1.3.2	Implement gateway payment and callback (F4, F23)	three-point	24	40	72	42.7	Chawanakorn S. 70 %, Watayut A. 30 %

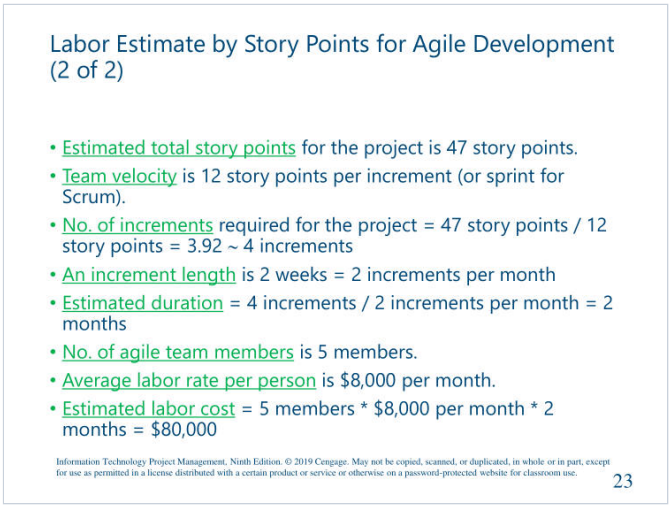


Figure 19. ch7 S23 · Labor estimate by user story points

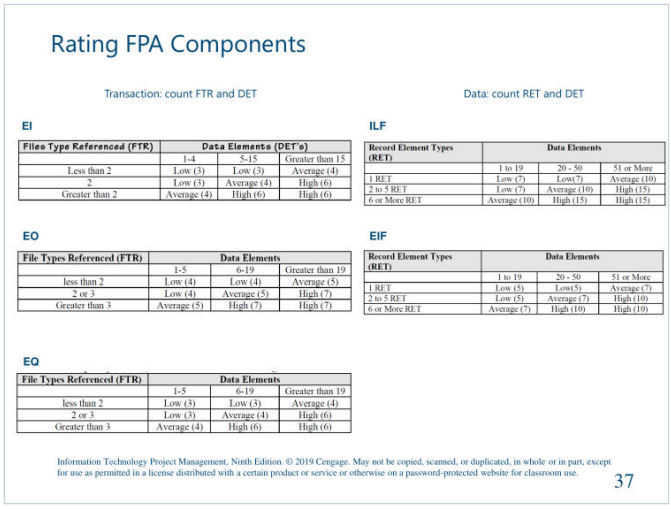


Figure 20. ch7 S37 · Function point complexity matrices

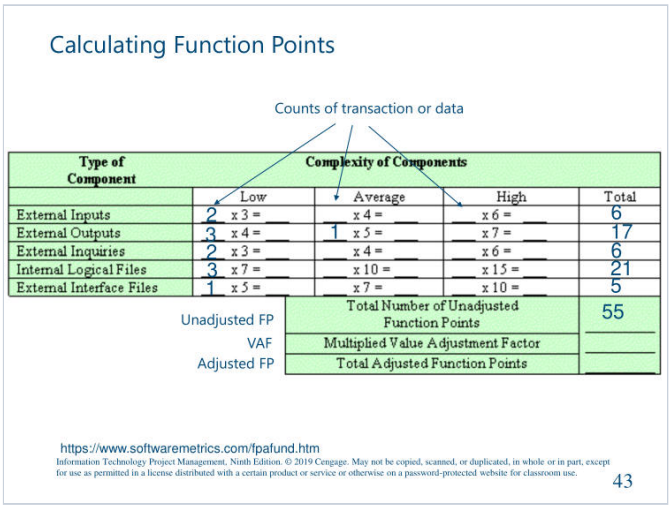


Figure 21. ch7 S43 · Function point worksheet

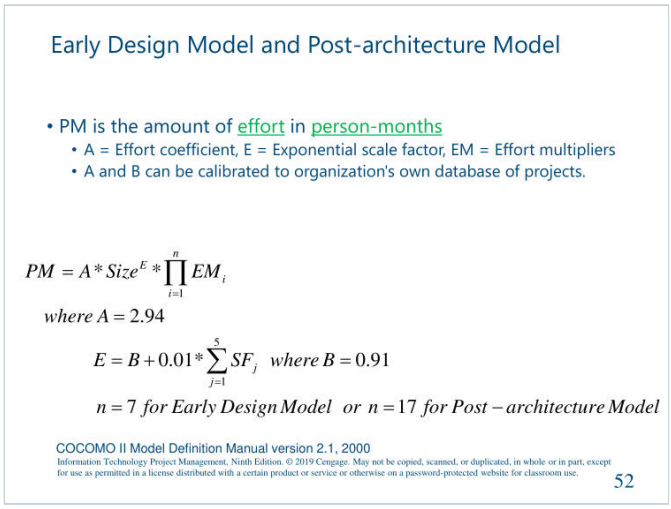


Figure 22. ch7 S52 · COCOMO II effort formula

Pet Care **never estimated effort**: every development task got the same 16 days, a login page and an AI model alike. Don't repeat that; it's the clearest place to score PMBOK-correctness points.

5. Human and physical resource assignment, resource rates

Concepts (ch9)

- Processes: plan resource management, estimate activity resources (**human and physical**), acquire, develop team, manage team, control resources (ch9 S23). Outputs include physical resource assignments, team assignments and resource calendars.
- **RAM** maps WBS work to people/OBS units; "for small projects, assign individual to WBS activities" (ch9 S28; PMBOK7 p.237 / PDF 342 "a grid that shows the project resources assigned to each work package").
- **RACI**: R does the task, A signs off (exactly one A per task), C consulted, I informed (ch9 S29).
- **Resource histogram**: number of resources over time (ch9 S30).
- **RBS** (resource breakdown structure): resources by category and type: facility, equipment, software, staffing (ch9 S33, Library Management System example).
- **Resource loading / overallocation**: "100 % = 8 hours per day"; above 100 % is overallocated (ch9 S36–37).
- **Resource leveling**: delay tasks within their slack to remove peaks, sometimes delaying the finish (ch9 S38–39, example 8/6/4 workers leveled to 6/6/6).
- Physical resources = "any resource that is not a person ... materials, equipment, software, testing environments, licenses" (PMBOK7 p.65 / PDF 160).
- The syllabus Workshop 3 (wk 12) and Workshop 4 (wk 13) cover "resource assignment – task type" and "–baseline" in the tool. Task types are **not in the slides**; in ProjectLibre/MS Project they are fixed units (default), fixed work and fixed duration, and decide which of work = duration × units changes when you edit one of them. Use **fixed work** for tasks whose effort you estimated, so adding a person shortens the duration rather than adding hours.

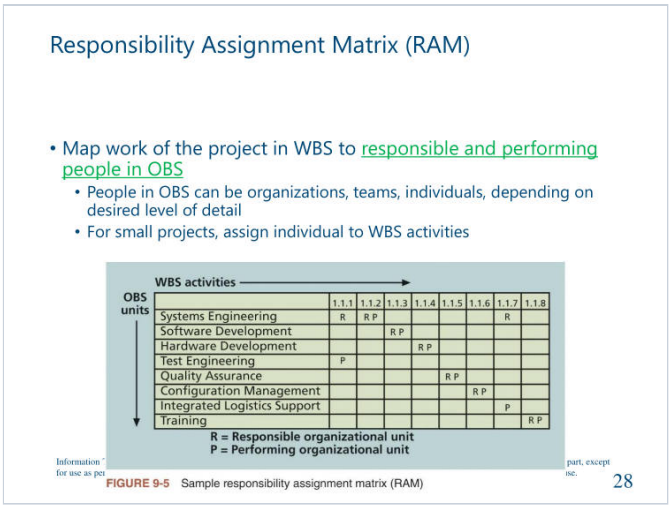


Figure 23. ch9 S28 · Responsibility assignment matrix (Figure 9-5)

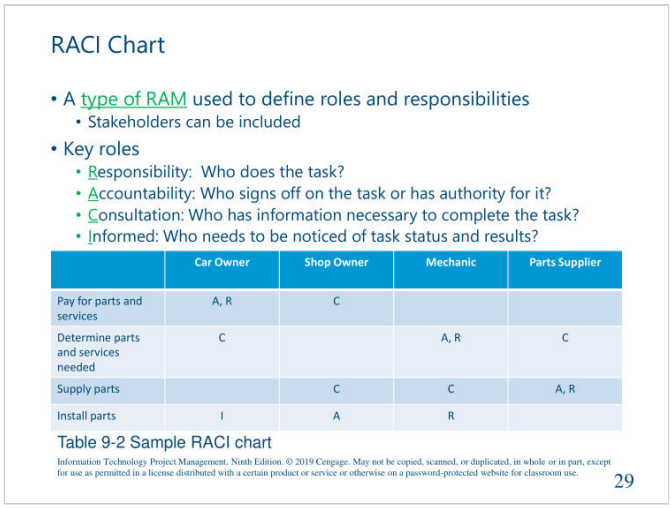


Figure 24. ch9 S29 · RACI chart (Table 9-2)

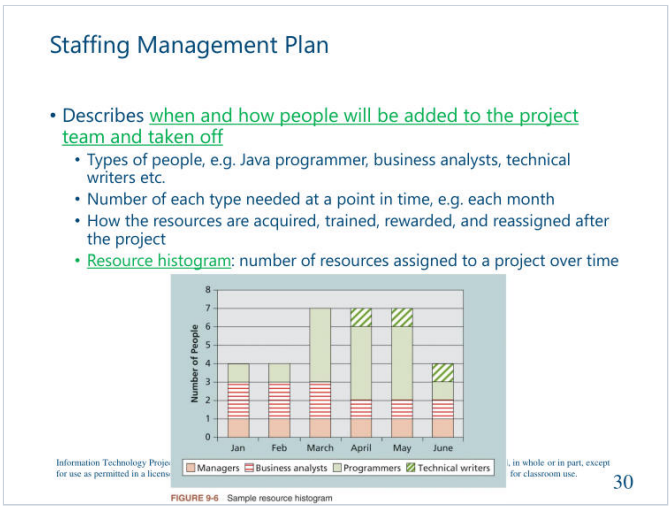


Figure 25. ch9 S30 · Staffing management plan and resource histogram (Figure 9-6)

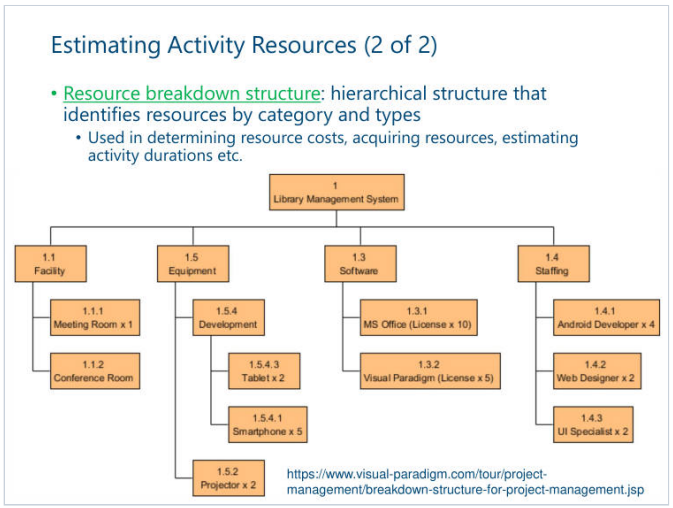


Figure 26. ch9 S33 · Resource breakdown structure (human and physical)

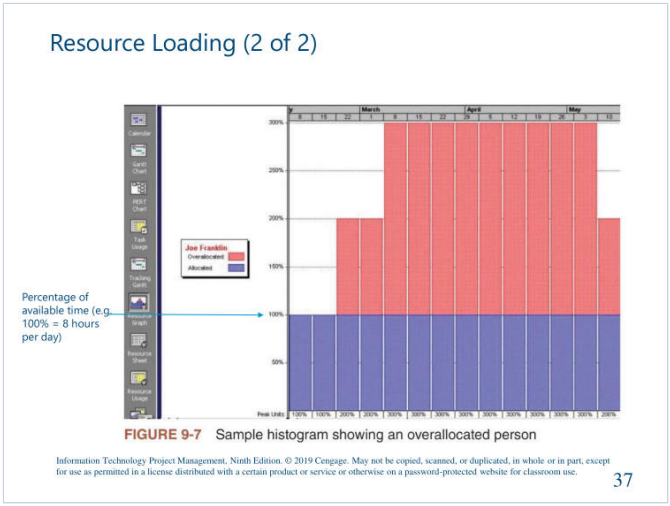


Figure 27. ch9 S37 · Resource loading and overallocation (Figure 9-7)

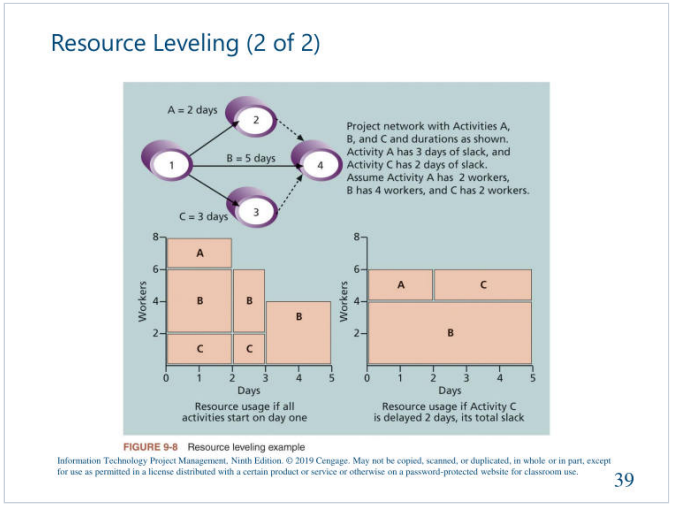


Figure 28. ch9 S39 · Resource leveling (Figure 9-8)

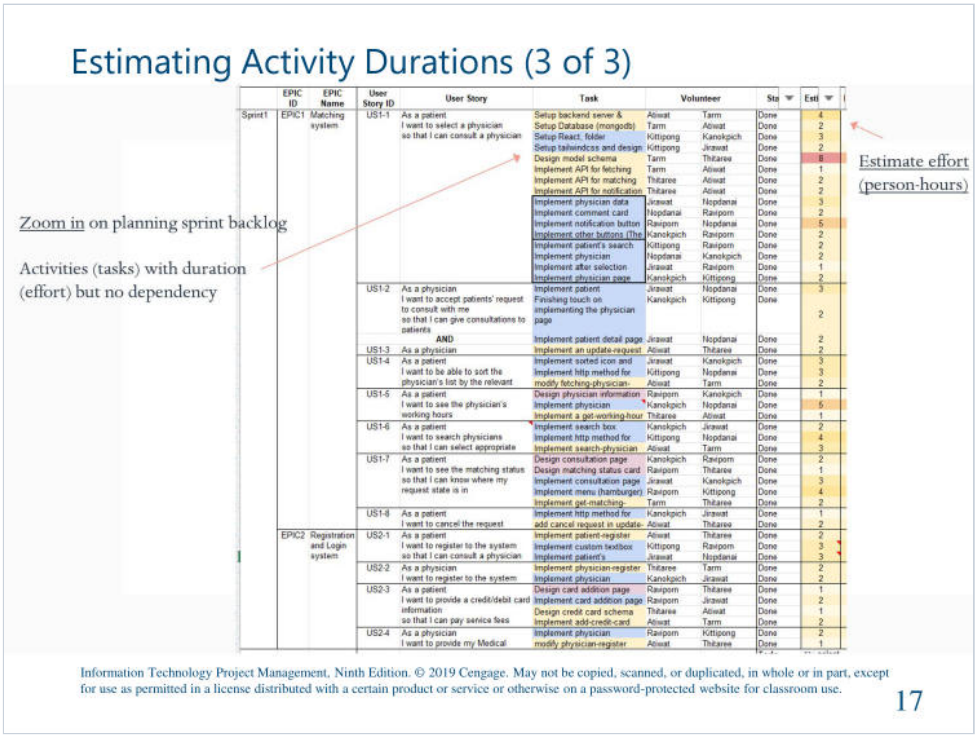


Figure 29. ch6 S17 · Sprint backlog: tasks assigned to members with effort in person-hours

For CTRS

Resource rate table (Pet Care Table 29 columns `No. | Resource Name | Cost (THB) | Unit`; improve with type and availability):

Resource	Type	Rate	Unit	Max units / availability
Boonnisa T. (PM)	Work	350	THB/h	part-time, h/week per phase from M1 Table 3.4 (e.g. 8 h/week in Inception = 20 %)
Sirapatr C. (SA/Architect)	Work	350	THB/h	…
Chawanakorn S. (Req. Spec./Designer/ Implementer)	Work	300	THB/h	…
Watayut A. (QA/Tester/Implementer)	Work	300	THB/h	…
Test devices (phone + tablet)	Material/ cost	15,000	per set, once	1
Dev/staging hosting	Cost	300	THB/ month	
Production hosting	Cost	700	THB/ month	
Domain + TLS	Cost	1,200	per year	
Free tiers (gateway sandbox, LINE Developers, GitHub, Figma, PlantUML)	Cost	0		
Legal / company set-up (Year 1)	Cost	15,000	once	

Keep the rates **per hour** and the same as M1 (Pet Care mixed per month / per day / per transaction and couldn't reconcile). Mention the resource calendar: part-time engineers, Thai public holidays and the New Year break (C1 ends 18 Dec, C2 starts 4 Jan).

Assignment to tasks: one table `WBS ID | Task | Resource (units %) | Work (h) | Physical resources` or a RAM/ RACI matrix (tasks × 4 members + Owner/Manager). Respect M1's rules: one owner per artifact and **the reviewer is never the author** (QA reviews). Make sure the role names match the rate table exactly (a Pet Care error). Then check the tool's **resource usage / histogram** for overallocation (> 100 % of the person's availability) and level it; show the histogram or say how you leveled.

Pet Care's assignment table (Table 30) only listed role names per task: no hours, no units, so it could not drive cost. They also copy-pasted task names (Sprints 6, 8, 11). Ours should carry hours and units.



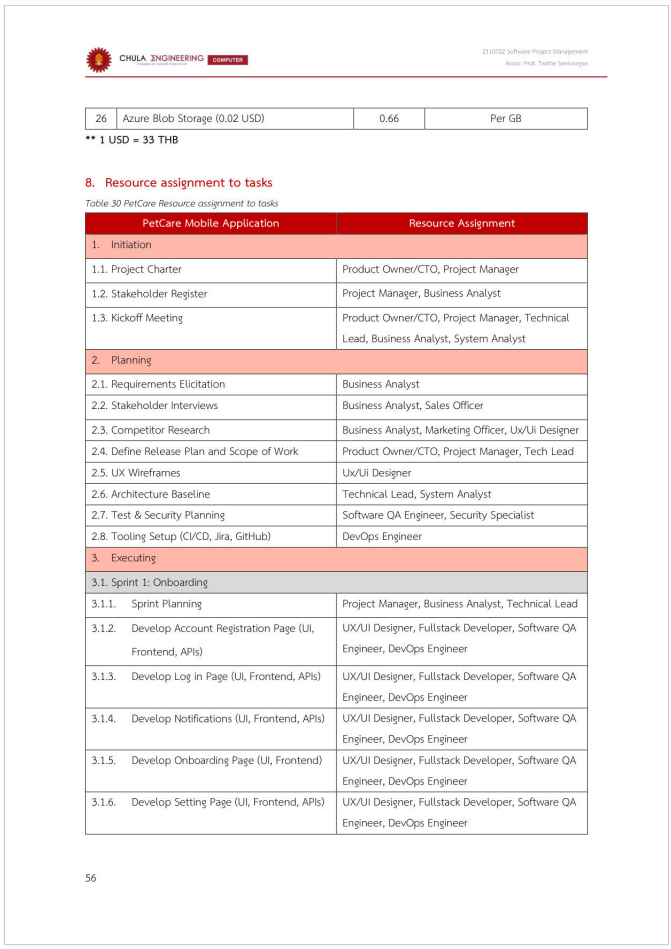
25. Create Project Documents
26. Release Candidate Build
27. Regression & Security Testing
28. Lessons Learned Workshop
29. Project Sign-Off

7. Resource rate

Table 29 PetCare Resource rate

No.	Resource Name	Cost (THB)	Unit
1	CTO/Product Owner (1)	120,000	Per Month
2	Project Manager (1)	70,000	Per Month
3	Technical Lead (1)	70,000	Per Month
4	Senior Business Analyst (1)	50,000	Per Month
5	Senior System Analyst (1)	50,000	Per Month
6	Senior UX/UI Designer (1)	40,000	Per Month
7	Senior Fullstack Developer (1)	55,000	Per Month
8	Junior Fullstack Developer (2)	30,000	Per Month
9	Senior Software Quality Assurance Engineer (1)	40,000	Per Month
10	Senior DevOps Engineer (1)	60,000	Per Month
11	Security Specialist (1) -> Calculate by job	10,000	Per Day (7 Days/Time/Year)
12	Data Scientists & AI Engineer (1)	70,000	Per Month
13	Marketing Officer (1)	30,000	Per Month
14	Sales Officer (1)	30,000	Per Month
15	Customer Support (1)	25,000	Per Month
16	Veterinary Advisor (1) -> Calculate by job	3,000	Per Day (10 Days/Time/Year)
17	Microsoft 365 (3 USD)	99	Per User Per Month
18	Github (21 USD)	693	Per User Per Month
19	Docker (15 USD)	495	Per User Per Month
20	Azure AI Vision (0.4 USD)	13.2	Per Transaction
21	Azure Kubernetes Service (0.1 USD) (3 Cluster)	3.3	Per Cluster Per Hour
22	Domain (1st Year 225.73/Year)	698.57	698.57 Per Year
23	Azure Monitoring (0.65 USD)	21.45	Per GB
24	Azure API Management (150.01 USD)	4,950.33	Per Month
25	Azure Key Vault – Secret Ops (0.03 USD)	0.99	Per 10,000 Transactions

Figure 30. Pet Care p.63 · Resource rate table (Table 29)



26	Azure Blob Storage (0.02 USD)	0.66	Per GB

** 1 USD = 33 THB

8. Resource assignment to tasks

Table 30 PetCare Resource assignment to tasks

PetCare Mobile Application	Resource Assignment
1. Initiation	
1.1. Project Charter	Product Owner/CTO, Project Manager
1.2. Stakeholder Register	Project Manager, Business Analyst
1.3. Kickoff Meeting	Product Owner/CTO, Project Manager, Technical Lead, Business Analyst, System Analyst
2. Planning	
2.1. Requirements Elicitation	Business Analyst
2.2. Stakeholder Interviews	Business Analyst, Sales Officer
2.3. Competitor Research	Business Analyst, Marketing Officer, Ux/Ui Designer
2.4. Define Release Plan and Scope of Work	Product Owner/CTO, Project Manager, Tech Lead
2.5. UX Wireframes	Ux/Ui Designer
2.6. Architecture Baseline	Technical Lead, System Analyst
2.7. Test & Security Planning	Software QA Engineer, Security Specialist
2.8. Tooling Setup (CI/CD, Jira, GitHub)	DevOps Engineer
3. Executing	
3.1. Sprint 1: Onboarding	
3.1.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.1.2. Develop Account Registration Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.3. Develop Log in Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.4. Develop Notifications (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.5. Develop Onboarding Page (UI, Frontend)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.6. Develop Setting Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer

Figure 31. Pet Care p.64 · Resource assignment to tasks (Table 30)

6. Cost estimation (labor, other costs, contingency reserve)

Concepts (ch7, PMBOK7)

- Direct vs indirect costs; PMs focus on **direct** costs; don't include sunk costs (ch7 S5).
- **Contingency reserve**: for **identified** risks ("known unknowns"), **included in the cost baseline**. **Management reserve**: for unidentified risks ("unknown unknowns"), needs higher-management approval, **not in the cost baseline** (ch7 S6; PMBOK7 p.127 / PDF 222, glossary PDF 332 and 337).
- Budget build-up (ch7 S71 Fig 7-5; PMBOK7 Fig 2-18 p.63 / PDF 158): `activity estimates + activity contingency → work package estimates → + contingency reserve → control accounts = cost baseline → + management reserve = project budget`.
- **Cost baseline** = "time-phased budget … to measure and monitor cost performance"; **project budget = cost baseline + management reserve** (ch7 S69). Cost management plan: rounding (e.g. nearest 100), units, contingency typically **10–20 %**, control threshold e.g. ±10 % (ch7 S9).

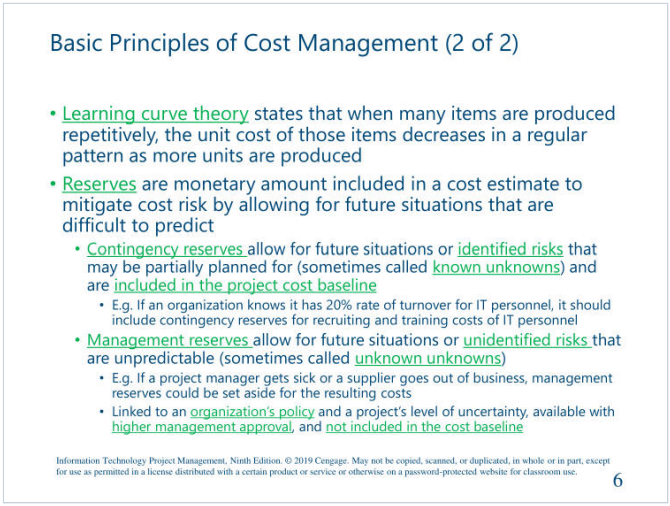


Figure 32. ch7 S6 · Contingency reserve vs management reserve

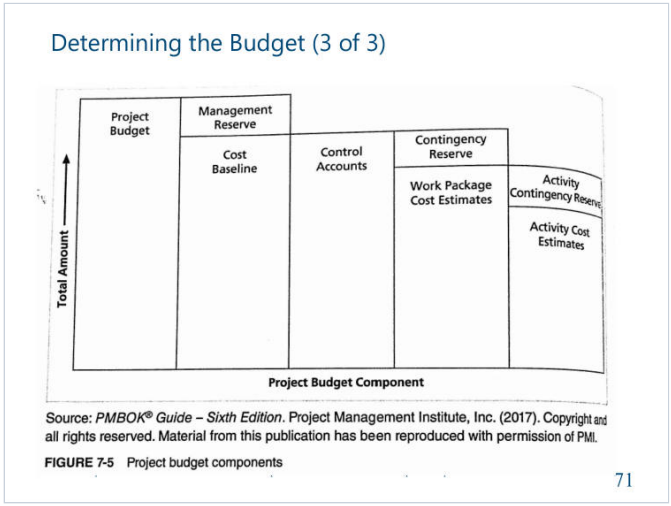


Figure 33. ch7 S71 · Budget components (Figure 7-5)

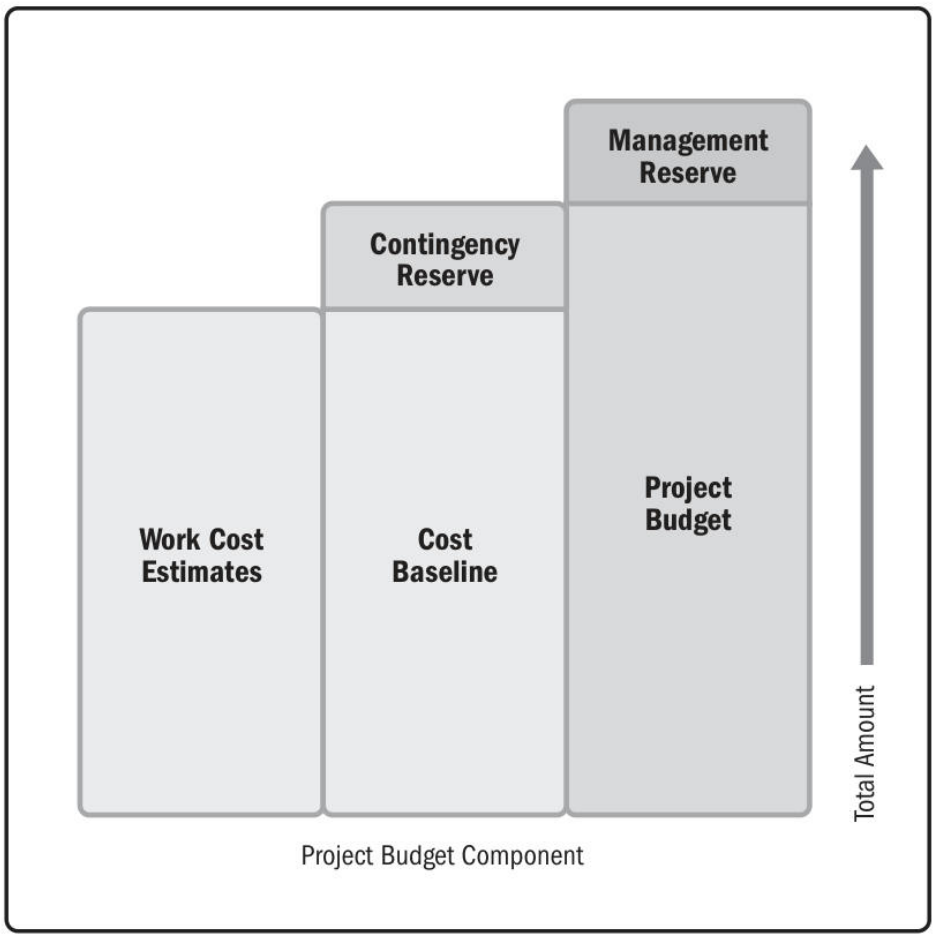


Figure 2-18. Budget Build Up

Figure 34. PMBOK7 Figure 2-18 · Budget build-up (p.63)

Model to copy: Surveyor Pro (ch7 S16, Figure 7-2)

Columns `WBS Items | # Units/Hrs. | Cost/Unit/Hr. | Subtotals | WBS Level 2 Totals | % of Total`, e.g. "Project manager 960 h × $100 = $96,000", "Servers 4 × $4,000", "Testing 10 % of hardware and software", "Reserves 20 % of total estimate", with the total. Then **Figure 7-4**: the same WBS items spread over months = time-phased cost baseline.

How to Develop a Cost Estimate and Basis of Estimates (2 of 3)

Surveyor Pro Project Cost Estimate Created October 5

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 2 Totals	% of Total
WBS Items					
1. Project Management				**$306,300**	**20%**
Project manager	960	$100	$96,000		
Project team members	1,920	$75	$144,000		
Contractors (10% of software development and testing)			$66,300		
2. Hardware				**$76,000**	**5%**
2.1 Handheld devices	100	$600	$60,000		
2.2 Servers	4	$4,000	$16,000		
3. Software				**$614,000**	**40%**
3.1 Licensed software	100	$200	$20,000		
3.2 Software development*			$594,000		
4. Testing (10% of total hardware and software costs)			$69,000	**$69,000**	**5%**
5. Training and Support				**$202,400**	**13%**
Trainee cost	100	$500	$50,000		
Travel cost	12	$700	$8,400		
Project team members	1,920	$75	$144,000		
Subtotal			$1,267,700		
6. Reserves (20% of total estimate)		0.2*1,267,700 →	$253,540	**$253,540**	**17%**
Total project cost estimate				**$1,521,240**	

*See software development estimate.

16

FIGURE 7-2 Surveyor Pro project cost estimate

Figure 35. ch7 S16 · Surveyor Pro cost estimate by WBS item (Figure 7-2)

Determining the Budget (2 of 3)

Surveyor Pro Project Cost Baseline Created October 10*

	Months												
WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
1.1 Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
1.2 Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
1.3 Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
5.1 Trainee cost									50,000				50,000
5.2 Travel cost									8,400				8,400
5.3 Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

*See the lecture slides for this chapter on the Instructor website for a larger view of this and other figures in this chapter. Numbers are rounded, so some totals appear to be off.

FIGURE 7-4 Surveyor Pro project cost baseline

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

70

Figure 36. ch7 S70 · Time-phased cost baseline (Figure 7-4)

For CTRS

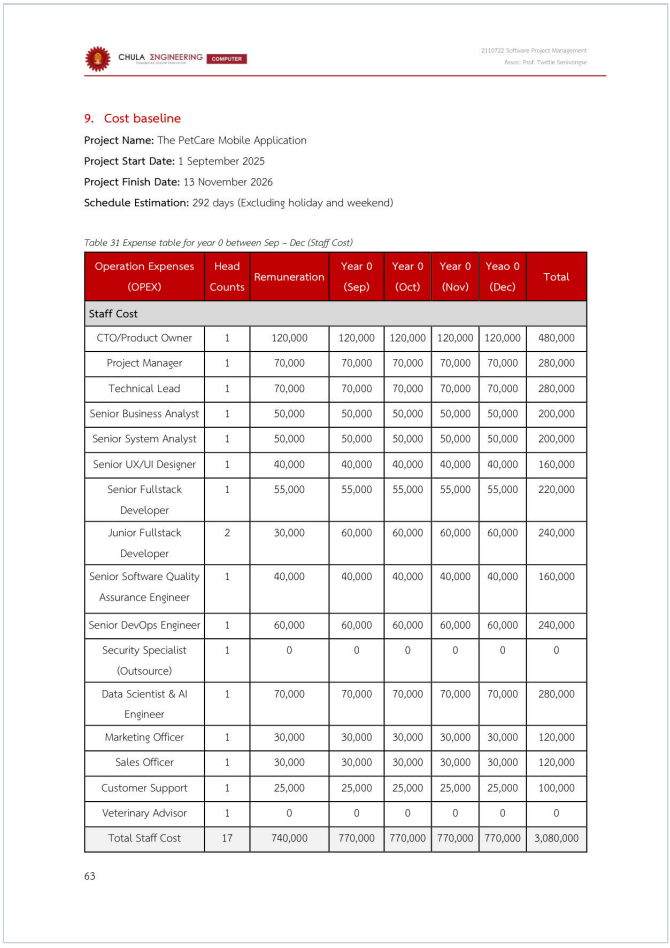
1. **Labor cost per task** = work hours × rate of each assigned resource. In the tool, the cost column is computed automatically. Roll up to work packages and level-2 WBS items.
2. **Other costs** by WBS item: devices 15,000 (Elaboration/C2 test), hosting by month, domain, legal 15,000 (Year 1), support of the pilot (8 h/month × 300 THB, Year 1; this is labor too).
3. **Contingency reserve**: tie it to the risk register, not just "10 %". E.g. RISK-02 part-time delay, RISK-03 gateway onboarding, RISK-05 load (an extra load-test cycle), RISK-08 late business rules (rework of R2.0). Either a per-risk amount (probability × impact cost) or a % justified by those risks. M1 used 10 % of Year 0 and of the productization work.

4. **Management reserve**: M1 had none. Decide as a team: either add one **outside** the baseline (and show baseline vs project budget per Fig 7-5) or state that the sponsor (the software house) holds none. Either way, say it explicitly; a grader will look for the contingency/management distinction.
5. **Time-phase it**: cost per month (or per iteration) and cumulative (S-curve), like Fig 7-4. The tool's cash-flow report can produce this.
6. **Reconcile**: total cost baseline vs M1 budget 1,010,590 THB; bottom-up hours vs top-down 2,632 h (percent difference). If different, explain why and record it (Revision History + Review Board).

Table layout suggestion:

WBS	Item	Hours / units	Rate	Labor	Other	Subtotal	%
1.1	Project Management	…					
1.4	Release 1.0 Core	…					
…	Contingency reserve (RISK-02, -03, -05, -08)						
	Cost baseline						
	Management reserve (if any)						
	Project budget						

Pet Care errors to avoid: costs not derived from assignments (headcount × salary × months), no reserve in the cost baseline although their business case had 15 %, a "Total" row summing unit prices, Year 1 charged 12 months although the project ended in November, rates in the cost table differing from the rate table.



9. Cost baseline

Project Name: The PetCare Mobile Application
Project Start Date: 1 September 2025
Project Finish Date: 13 November 2026
Schedule Estimation: 292 days (Excluding holiday and weekend)

Table 31 Expense table for year 0 between Sep – Dec (Staff Cost)

Operation Expenses (OPEX)	Head Counts	Remuneration	Year 0 (Sep)	Year 0 (Oct)	Year 0 (Nov)	Yeao 0 (Dec)	Total
Staff Cost							
CTO/Product Owner	1	120,000	120,000	120,000	120,000	120,000	480,000
Project Manager	1	70,000	70,000	70,000	70,000	70,000	280,000
Technical Lead	1	70,000	70,000	70,000	70,000	70,000	280,000
Senior Business Analyst	1	50,000	50,000	50,000	50,000	50,000	200,000
Senior System Analyst	1	50,000	50,000	50,000	50,000	50,000	200,000
Senior UX/UI Designer	1	40,000	40,000	40,000	40,000	40,000	160,000
Senior Fullstack Developer	1	55,000	55,000	55,000	55,000	55,000	220,000
Junior Fullstack Developer	2	30,000	60,000	60,000	60,000	60,000	240,000
Senior Software Quality Assurance Engineer	1	40,000	40,000	40,000	40,000	40,000	160,000
Senior DevOps Engineer	1	60,000	60,000	60,000	60,000	60,000	240,000
Security Specialist (Outsource)	1	0	0	0	0	0	0
Data Scientist & AI Engineer	1	70,000	70,000	70,000	70,000	70,000	280,000
Marketing Officer	1	30,000	30,000	30,000	30,000	30,000	120,000
Sales Officer	1	30,000	30,000	30,000	30,000	30,000	120,000
Customer Support	1	25,000	25,000	25,000	25,000	25,000	100,000
Veterinary Advisor	1	0	0	0	0	0	0
Total Staff Cost	17	740,000	770,000	770,000	770,000	770,000	3,080,000

63

Figure 37. Pet Care p.71 · Staff cost, Year 0 (Table 31)

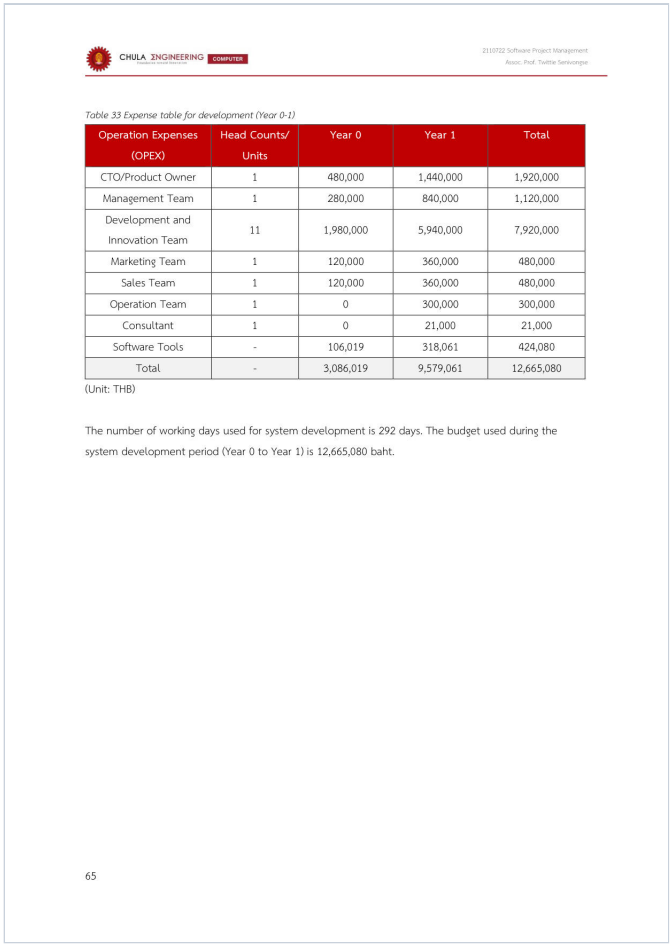


Table 33 Expense table for development (Year 0-1)

Operation Expenses (OPEX)	Head Counts/ Units	Year 0	Year 1	Total
CTO/Product Owner	1	480,000	1,440,000	1,920,000
Management Team	1	280,000	840,000	1,120,000
Development and Innovation Team	11	1,980,000	5,940,000	7,920,000
Marketing Team	1	120,000	360,000	480,000
Sales Team	1	120,000	360,000	480,000
Operation Team	1	0	300,000	300,000
Consultant	1	0	21,000	21,000
Software Tools	-	106,019	318,061	424,080
Total	-	3,086,019	9,579,061	12,665,080

(Unit: THB)

The number of working days used for system development is 292 days. The budget used during the system development period (Year 0 to Year 1) is 12,665,080 baht.

65

Figure 38. Pet Care p.73 · Development cost Years 0–1 (Table 33)

7. Schedule: Gantt chart and critical path

Concepts (ch6)

- Processes: define activities → sequence → estimate durations → develop schedule → control (ch6 S5–6). Output: **schedule baseline** = "entire approved planned schedule" (ch6 S23–25).
- **Dependencies** (ch6 S10): mandatory (hard logic), discretionary (soft logic, use with care), external (e.g. gateway merchant approval, LINE OA settings, La Loy's workshop), internal.
- **Relationships** (ch6 S14): FS (most common), SS, FF, SF. **Lead** = successor starts before the predecessor finishes (FS − 5 days); **lag** = a wait (FS + 3 days) (ch6 S8).
- **Gantt chart symbols** (ch6 S19): diamonds = milestones, thick bars = summary tasks, light bars = durations, arrows = dependencies. Gantt with resource names on the bars (ch6 S21).
- **Critical path** = "the longest path through the project", which determines the shortest possible duration; activities with the least (zero) slack (ch6 S26; PMBOK7 glossary PDF 333).
- **Slack/float**: total slack = delay without delaying the project finish; free slack = without delaying successors' early start (ch6 S30–32). (PMBOK7 doesn't define float; cite the slides.)
- **Forward / backward pass** (ch6 S30–32): EF = ES + dur, ES = max EF of predecessors; LS = LF − dur, LF = min LS of successors; **TS = LS − ES = LF − EF**; FS = min ES(successors) − EF. Worked example Project X: paths 14, **16 (B-E-H-J, critical)**, 9, 14 (ch6 S27, slacks in Table 6-1 S34).
- **Schedule compression** (ch6 S38): **crashing** (add resources for the least extra cost: full time instead of part time, a temporary hire) and **fast tracking** (overlap tasks). Use these if the critical path misses a charter date, and say so.
- **Critical chain** (ch6 S40–46): considers resource constraints too. Relevant for us because the same four people do everything: the resource-leveled schedule may have a longer critical chain than the logic-only critical path. Optional but a good discussion point.
- Discipline (ch6 S48): review the draft schedule in the charter, detail it with the team, get approval, keep it realistic.

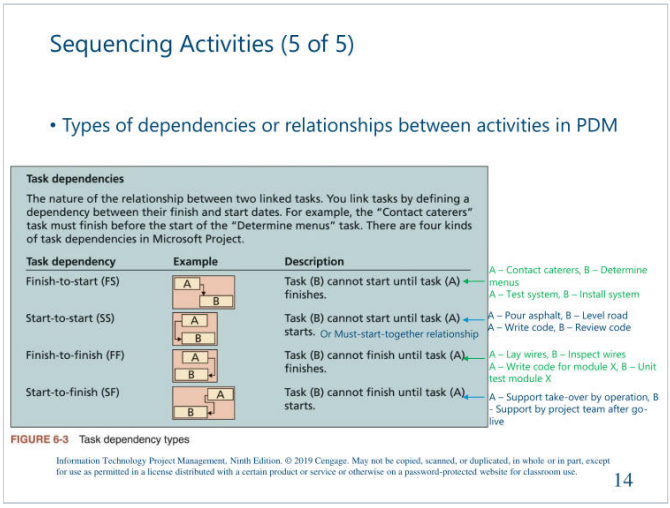


Figure 39. ch6 S14 · Task dependency types FS, SS, FF, SF (Figure 6-3)

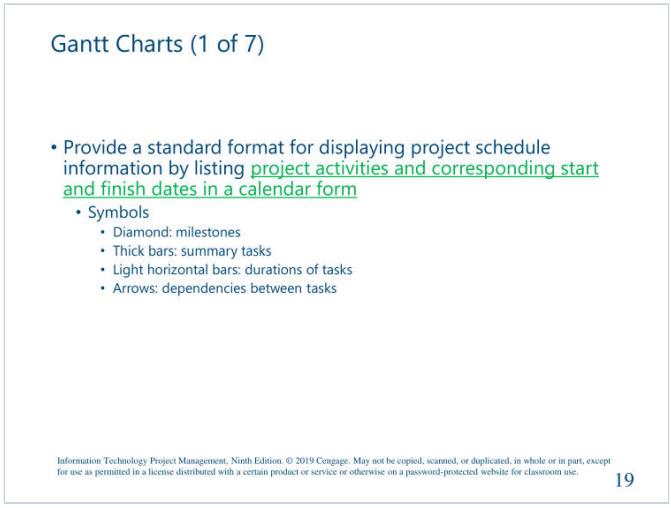


Figure 40. ch6 S19 · Gantt chart symbols

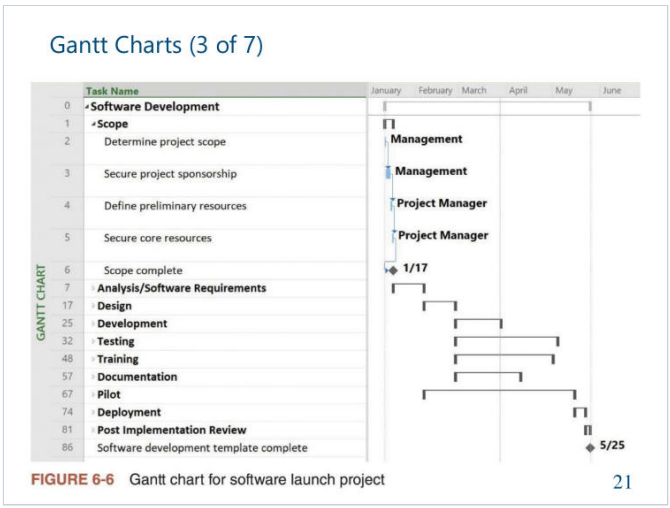


Figure 41. ch6 S21 · Gantt chart with resources and milestones (Figure 6-6)

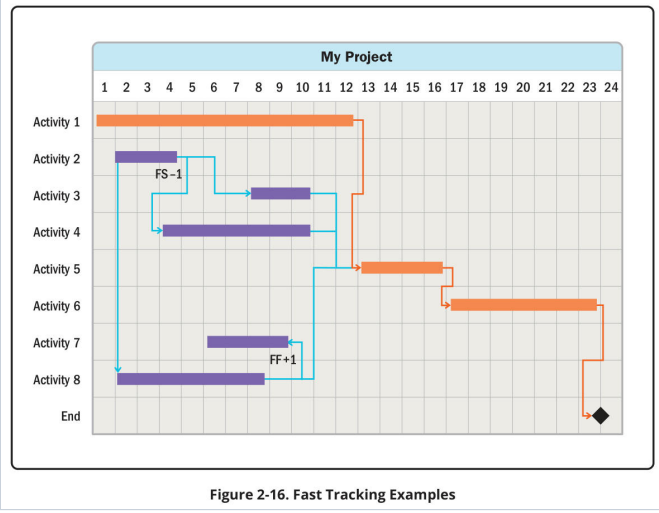


Figure 42. PMBOK7 Figure 2-16 · Fast tracking, leads and lags (p.60)

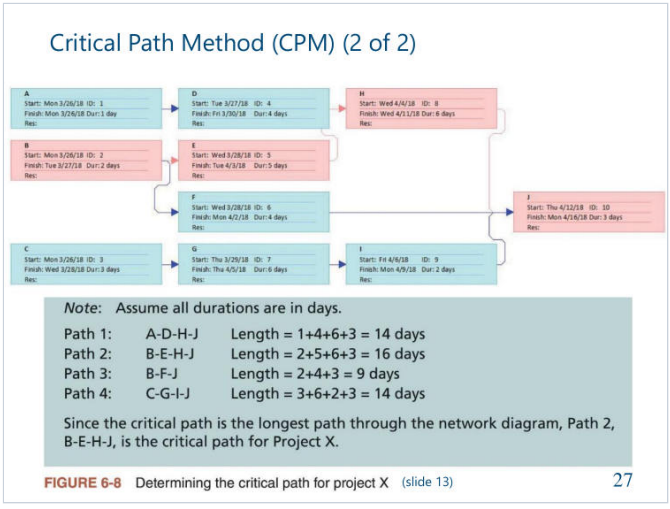


Figure 43. ch6 S27 · Critical path of Project X (Figure 6-8)

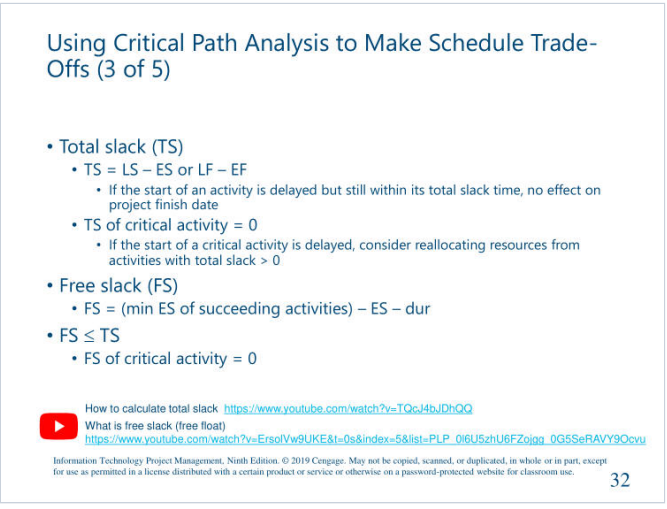


Figure 44. ch6 S32 · Forward/backward pass and slack rules

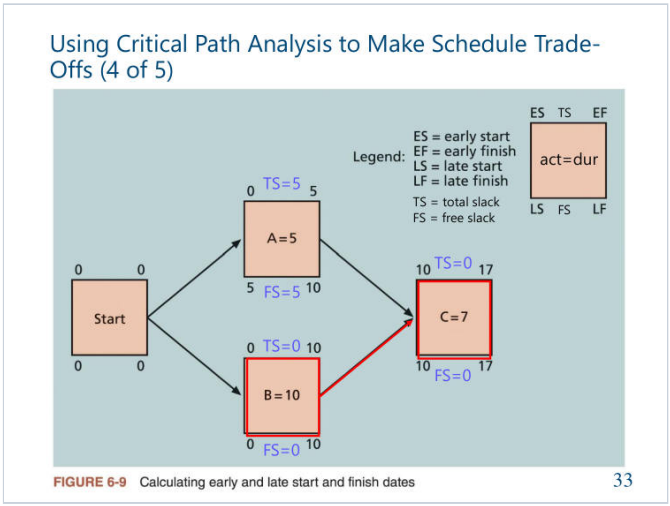


Figure 45. ch6 S33 · Forward/backward pass worked example (Figure 6-9)

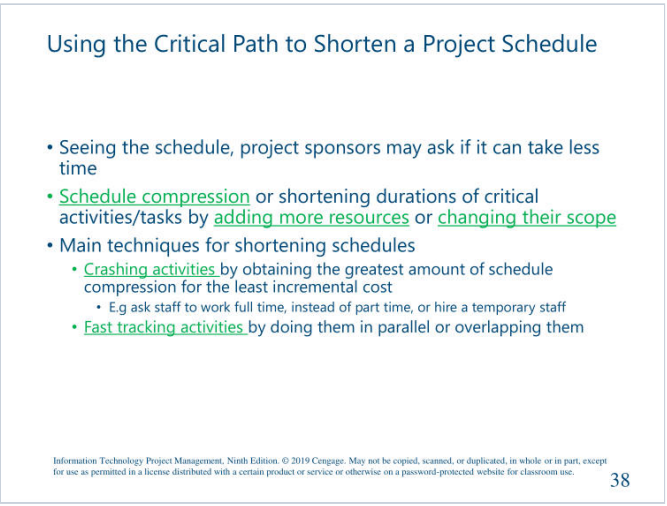


Figure 46. ch6 S38 · Crashing and fast tracking

For CTRS

1. Build it in **ProjectLibre** from the WBS of §3: tasks with work hours, resources with max units, dependencies (FS mostly; SS for e.g. "write test cases" alongside "write use-case spec"; external dependency for gateway onboarding before LCA), and **0-day milestones** on the M1 dates.

2. **Hit the M1 dates** (Table 2.21). If the bottom-up estimate doesn't fit an iteration, resolve it by leveling, crashing (more hours/week), fast tracking or moving priority-2 features (M1 RISK-02 response), and record that decision.
3. Present (Pet Care did 1 and 2 well, copy that): - Gantt screenshot (summary level readable in the PDF; full detail in an appendix or linked file); - task table `WBS | Task | Duration | Start | Finish | Predecessors | Resources`; - **critical path**: highlight it in the Gantt (ProjectLibre marks critical tasks in red) and list it, ideally with total slack per task (Table 6-1 style). Make sure listed tasks really have 0 slack (Pet Care listed tasks with slack and contradicted its own text).
4. Remove tool artefacts (the "?" estimated-duration flags Pet Care left in) and keep one duration count (they had 292 vs 315 days).
5. Long project (2 years) + iterative RUP: plan near iterations in detail and later ones (C4, T3, P1) at a higher level. That is **rolling wave planning**, a PMBOK term (PMBOK7 glossary PDF 344) worth naming.

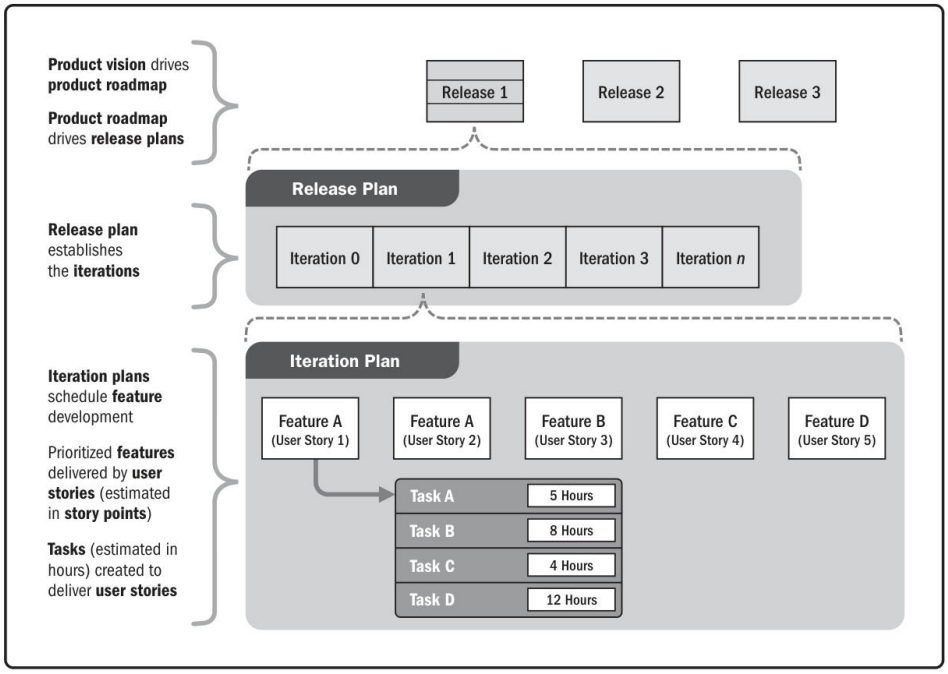


Figure 2-17. Release and Iteration Plan

Figure 47. PMBOK7 Figure 2-17 · Release and iteration plan (p.61)

The existing M1 Gantt source is `assets/diagrams/schedule-gantt.puml` (phase level); M2 needs the task level from the tool.

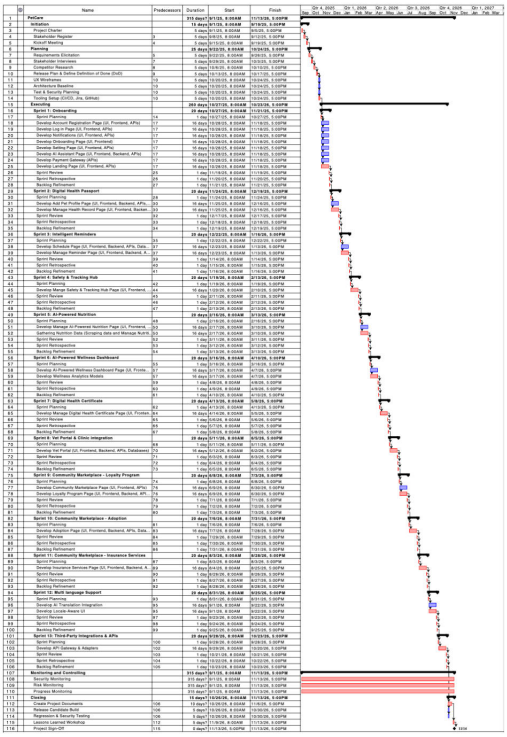


Figure 7 PetCare Gantt chart

Figure 48. Pet Care p.61 · Gantt chart from ProjectLibre (Figure 7)

6.2. Critical path

From the analysis of the Critical Path in the PetCare Application Project, which follows the Agile Framework, it was found that the sequence of critical activities represents essential tasks that must be executed continuously from project initiation (September 2025) through Phase 2 completion (31 July 2026) for proof-of-concept delivery. Phase 3 (1 Aug 2026 – 13 Nov 2026) execution is conditional on Phase 2 go/no-go decision. Any delay in these activities will directly impact the overall project timeline, potentially extending the completion date and affecting the delivery schedule and development milestones.

1. Project Charter
2. Stakeholder Register
3. Kickoff Meeting
4. Requirements Elicitation
5. Stakeholder Interviews
6. Competitor Research
7. Define Release Plan and Scope of Work
8. UX Wireframes
9. Architecture Baseline
10. Test & Security Planning
11. Tooling Setup (CI/CD, Jira, GitHub)
12. Sprint 1
13. Sprint 2
14. Sprint 3
15. Sprint 4
16. Sprint 5
17. Sprint 6
18. Sprint 7
19. Sprint 8
20. Sprint 9
21. Sprint 10
22. Sprint 11
23. Sprint 12
24. Sprint 13

Figure 49. Pet Care p.62 · Critical path as a list (section 6.2)

8. Pet Care (2025) at a glance: copy vs avoid

Copy	Avoid
One thread from business case → charter → WBS → Gantt → cost; same names everywhere	WBS numbering broken (5.2.1.x, 6.x, 3.x, 1.10.x) and different again in the assignment table
Numbered captions, list of tables/figures, version history	No effort estimation; every dev task 16 days
Scope statement in the textbook template, KPIs as acceptance criteria	Template text left in; filler justification
Gantt as screenshot **and** table with predecessors and dates, plus a link to the source file	Critical path asserted, includes tasks with slack, contradicts text; 292 vs 315 days
WBS dictionary fields: input, output, acceptance criteria, responsibility	Dictionary only for PM items; no effort/cost/predecessors
Rate table with source currency and exchange rate	Mixed rate units; role names that don't match; copy-paste task names
	Cost not built from assignments; no contingency/management reserve in baseline; totals that don't add up
	No Review Board log, no GenAI log (required for us)

Section layout they used (PDF pages): 5 Scope Baseline (5.1 statement, 5.2 WBS, 5.3 dictionary) p.53–60 · 6 Schedule Baseline (6.1 Gantt, 6.2 critical path) p.60–63 · 7 Resource rate p.63–64 · 8 Resource assignment p.64–70 · 9 Cost baseline p.71–73.

9. GenAI log (Part 1) and prompting

- **8 records** (2 per member), each on a new page, fields: Artifact, Member, AI Tool, Initial Prompt, Initial Output, Review (against Requirement Satisfaction / PMBOK Correctness / Project Consistency), Revision, Final Output (brief, M2 section). Same format as M1's `part1-genai-log.md`.
- Use **CREATE** for complex artifacts (WBS, estimates, cost) and **RTF** for quick ones; label each part, e.g. `[Character] You are a senior project manager…`.
- Give the AI the **single source of truth** every time: charter, release/feature table, M1 Table 2.21 (dates), Table 3.4 (team, hours, rates), risks. Most "Project Consistency" failures come from the AI inventing dates, roles or rates.
- AI is good at: drafting the scope statement, a first WBS decomposition, WBS dictionary entries, three-point estimates to challenge, a cost table layout. AI is bad at: arithmetic over many rows, critical path, consistency across tables. **Compute those in the tool or a spreadsheet** and say so in the Revision field; that is the "human-in-the-loop" the brief grades.
- Suggested split (2 prompts each), matching M1 roles:

Member	Prompt 1	Prompt 2
Sirapatr C. (SA)	Scope statement: summary, product characteristics, exclusions	WBS dictionary entries
Chawanakorn S. (Req./ Designer)	WBS and task decomposition of Construction (features → tasks)	Effort estimation (three-point / FP or story points)
Watayut A. (QA)	Effort cross-check and estimate review (percent difference vs M1)	Cost estimation, contingency reserve from risks
Boonnisa T. (PM)	Resource assignment, rates, leveling	Schedule: dependencies, milestones, critical path review

Example skeleton (CREATE):

```
[Character] You are a senior software project manager experienced with PMBOK and RUP.
[Request] Decompose work package 1.4.1 "Construction C1" of the CTRS project into activities (verbs) with
person-hour estimates (optimistic, most likely, pessimistic).
[Examples] "1.4.1.3.2 Implement gateway payment and callback (F4, F23) | O 24 | M 40 | P 72 | Chawanakorn
S."
[Adjustments] Features of C1: … (paste). Team and rates: … (paste Table 3.4). C1 runs 26 Oct–18 Dec 2026;
the reviewer of an artifact is never its author; hours must fit each member's availability.
[Type of Output] Markdown table: WBS ID | Activity | Feature IDs | O | M | P | PERT (O+4M+P)/6 | Resource.
[Evaluation] Every C1 feature covered exactly once (100 % rule); nouns for work packages, verbs for
activities; total within ±15 % of the M1 C1 share of Table 3.4.
```

10. Checklist before the Review Board

- [ ] One WBS numbering used in the WBS, dictionary, assignment, cost and Gantt tables
- [ ] Every feature F1–F35 and Release 2.1 item appears in exactly one work package; F36/F37 listed as excluded
- [ ] Work packages are nouns, activities are verbs; milestones are 0-day with M1 dates
- [ ] Effort in person-hours per task, method stated, two approaches compared (percent difference)
- [ ] Every task has a resource with units; no resource over 100 % of availability (or leveling explained)
- [ ] Rates per hour identical to M1; role names identical everywhere
- [ ] Labor cost = hours × rate (from the tool); other costs listed; contingency reserve tied to RISK-nn; management reserve decision stated
- [ ] Cost baseline time-phased; totals add up; reconciled with M1 budget 1,010,590 THB (or variance explained)
- [ ] Gantt shows milestones and critical path; listed critical tasks really have zero total slack; finish dates match M1 milestones
- [ ] Captions numbered; Document Revision History v2.0 row; Review Board log; 8 GenAI records
- [ ] Concise: summary tables in the body, long tables in an appendix or linked tool file